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Learning in speculative bubbles: Theory and experiment☆

Jieying Hong[a], Sophie Moinas[b], Sébastien Pouget[b,*]

[a] School of Economics and Management, Beihang University, 37 Xueyuan Road, Haidian District, Beijing, China
[b] University of Toulouse - Capitole, Toulouse School of Economics, TSM-Research, 1 Esplanade de l'Université, Toulouse Cedex 06, France

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ABSTRACT

Does learning reduce or fuel speculative bubbles? We study this issue in the context of the Bubble Game proposed by Moinas and Pouget (2013). Our theoretical analysis based on adaptive learning shows that i) in the long run, learning induces convergence to the unique no-bubble equilibrium, ii) in the short run, more experienced traders create more bubbles, and iii) learning is more difficult when more steps of reasoning are necessary to reach equilibrium. These predictions are consistent with our experimental observations. We find that reinforcement learning rather than belief-based learning is driving behavior in our experiment.

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1. Introduction

Do traders learn to avoid participating in speculative bubbles? Instead, does learning fuel speculative bubbles? These questions are crucial to determine whether macroeconomic policies are warranted to control the emergence and development of speculative bubbles. They have been the long-standing focus of the financial economics literature. On the one hand, Smith et al. (1988) propose an experimental design to study speculative bubbles and show that bubbles are less likely but do not disappear with experience. This result is confirmed by King et al. (1993). Dufwenberg et al. (2005) further show that bubbles also diminish with experience when only part of the traders are experienced. On the other hand, more recent papers suggest that bubbles can be rekindled when experienced traders are confronted with new market parameters (Hussam et al., 2008) and when new traders enter the market in an overlapping-generation experiment (Deck et al., 2014). The objective of this paper is to study the dynamics and the mechanisms of learning in speculative bubbles and to identify, both theoretically and experimentally, the conditions under which experience can be expected to shut down speculation.

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* Corresponding author.
E-mail addresses: sebastien.pouget@tse-fr.eu, spouget@univ-tlse1.fr (S. Pouget).

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As emphasized by Hussam et al. (2008), the experimental bubble literature lacks "a theory of agent adaption other than the backward-induction-optimal-decision approach or deviations theoreof", which hinders our understanding at the individual level on how traders think and adapt with experience. This may be largely due to the complexity of Smith et al. (1988)'s experimental environment with endogenous prices, which makes it difficult to consider adaptive learning models in this framework. To overcome this difficulty, we consider the bubble game designed by Moinas and Pouget (2013). This enables us to provide one of the first studies of speculative bubbles that focuses on the dynamics and the mechanisms of individual learning (see, also, the papers by Duffy and Ochs (1999) and Duffy (2001) that we discuss in the literature review section). Specifically, our paper not only provides a theory to capture traders' learning process but also empirically identifies both the existence of learning and the drivers of learning at work in our experiment.

In the bubble game, the trading asset has zero fundamental value, which is common knowledge to all market participants. Trading proceeds sequentially, traders' positions in the sequence are random but unobservable, and prices are exogenous with powers of 10 (prices may be equal to 1, 10, 100...). The first price, i.e., the trading price for the trader in the first position, is randomly determined, and the price for the trader in the next position is ten times the previous price.[1] In the game with a price cap, as shown by Moinas and Pouget (2013), there is no bubble at the dominance-solvable Bayesian Nash equilibrium: confronted with the highest potential price, a rational trader knows he is last in the sequence and refuses to buy. This decision requires no iterated reasoning. Anticipating this behavior, a rational trader receiving the second highest price should also refuse to buy. This decision requires one step of iterated reasoning. Proceeding backward, this logic, in spirit of Morris et al. (1995), rules out the formation of bubbles. The higher the price cap, the higher is the number of steps of iterated reasoning needed to reach equilibrium, i.e., to realize that not speculating is optimal. However, bubbles can emerge when traders suffer from bounded rationality (in the form of limited depth of reasoning as in cognitive hierarchy models, see, e.g., Camerer et al., 2004, or in the form of random mistakes as in the quantal response models, see, e.g., McKelvey and Palfrey, 1995).

The bubble game is useful to study the dynamics and mechanisms of learning in speculative episodes for four reasons. First, there is a finite number of decisions individuals can make. Second, there is a finite number of past events individuals can observe that may influence their current decision. These two features make it possible to use learning models to analyze the learning process of traders without making strong auxiliary assumptions. Third, the bubble game enables to study different types of speculative behavior: speculation that is due to costly mistakes and speculation that arises when traders try to profitably trade on others' mistakes. Fourth, the bubble game is short enough so that it can be repeated a large number of times. This enables to provide subjects with numerous learning opportunities.

Our experiment features several repetitions of the bubble game in a stranger design with anonymity. In our experiment, 132 subjects participate in a three-player bubble experiment similar to Moinas and Pouget (2013) with a two-by-two design: the cap on the first price is either 1 or 10,000 (corresponding to a maximum number of steps of reasoning of 2 and 6, respectively) and the number of replications of the experiment is either 10 or 20.

Our experimental data analyses reveal that learning may impede or foster speculative bubbles, depending on factors such as the number of steps of reasoning and the probability to be last in the market sequence. On the one hand, when the cap on the first price is 1, there are few steps of reasoning and the propensity to participate in bubbles tends to decrease slowly over time; when the cap is 10,000, there are many steps of reasoning and the propensity to speculate tends to increase. On the other hand, traders with a high probability to be last in a market sequence gradually learn not to speculate while other traders may learn to speculate more. Overall, our experimental results, at the market and individual levels, suggest that learning may sometimes translate into a higher propensity to speculate.[2]

To understand at the individual level how traders learn, adapt and make their trading decisions across replications, we capture traders' learning process in the bubble game using Camerer and Ho (1999)'s Experience-Weighted Attraction (EWA) model. Traders' choices depend on the various actions' attractions, which are updated according to actions' past performance. There are two key parameters in the model, namely the imagination parameter and the responsiveness parameter. When the imagination parameter is equal to 0, agents only reinforce actually chosen actions, as is the case in reinforcement learning. When the imagination parameter is greater than 0, agents can imagine and reinforce actions that were not actually chosen, as is implicitly assumed in belief-based learning. Camerer and Ho (1999)'s adaptive learning model is thus general in the sense that it nests belief-based learning and reinforcement learning. For a given trader, actions' attractions are transformed into choice probabilities via a logistic function with a given payoff responsiveness parameter. When this responsiveness parameter is 0, agents choose each action with the same probability, while when it is infinite, they choose with probability one the action with the highest attraction.

We show that in the long term, the market converges to the unique dominance-solvable no-bubble equilibrium. This is in line with the analysis of Milgrom and Roberts (1991). In the bubble game, the trader confronted with the highest price would only loose money by speculating and thus will learn to reduce his probability to buy. Once this probability

[1] The bubble game features a simple experimental set up. It constitutes a metaphor of naturally-occurring financial markets in which speculation can be clearly identified and studied. Additional features of these markets such as the existence of assets with risky cash flows, the presence of more traders, and endogenous trading sequences and prices, could be interesting to introduce in this set up. This would enable to test whether the basic speculative forces present in the simple bubble game are robust. This is left for future research.

[2] We also find that traders confronted with the highest potential price sometimes enter into bubbles. This indicates that speculation in our experiment is partly due to traders' mistakes. This is in line with the findings of Lei et al. (2001) obtained in the Smith et al. (1988)'s environment.

is sufficiently low, the trader with the second highest price will face the same situation and also see his probability to buy decrease. Eventually, all traders will learn to stop speculating. We also show that, in the short term, the propensity to speculate can increase for traders who are not confronted with the highest potential price. In order to study speculation in the intermediate term, we run simulations of the bubble game with 1000 independent trials that each includes 1000 successive runs. We find that learning initially increases traders' propensity to speculate and that it may take multiple periods before the propensity to speculate starts decreasing. More experienced traders thus initially create more bubbles. This is because traders who are not last are likely to gain from other traders' mistakes, especially at initial stages when other traders have limited experience. We also find that learning is quicker when the number of steps of reasoning is lower. Overall, our theoretical results are thus consistent with our experimental observations.

Additional empirical analyses aim at identifying the main type of learning at work in our experiment. To achieve this objective, we have performed both regression analyses and structural estimations. First, our regressions show that subjects' behavior changes depending on the outcomes of past actions. This is an evidence of adaptive learning at the individual level. In addition, we also find that subjects who had bought (and lost) when they had observed the maximum price tend to buy significantly less, which provides evidence of reinforcement learning. However, we find no evidence for belief-based learning since no significant effect was found on the probability to speculate for subjects who had not bought when offered the maximum price. Second, we structurally estimate the adaptive learning model on our experimental data. We find an imagination parameter of 0.45, not statistically different from zero. This indicates that imagination is not an important ingredient for learning in our experimental setting. Our estimate of the responsiveness parameter is 0.2. This suggests that there is a lot of noise in individual behavior. Vuong tests indicate that the learning model fits the data better than a no-learning model. Overall, our empirical analyses suggest that learning is at work in our experiment despite speculative bubbles frequently arising and that reinforcement is the main learning mechanism in our setting.

The rest of the paper is organized as follows. Next section offers a review of the related literature. Section 3 presents the bubble game and the experimental setup. Section 4 presents the experimental results. Section 5 offers an adaptive learning model of bubbles and a test of the model based on our experimental data. Section 6 concludes.

2. Related literature

In their seminal contribution, Smith et al. (1988) propose an experimental setup in which (irrational) bubbles can be studied, and they show that such bubbles arise in their experiments. This contribution has triggered an extremely large number of experimental and theoretical work on bubble formation (see for example, Deck et al. (2014) for a discussion of a number of the follow-up studies to Smith et al. (1988), and Caginalp and Ermentrout (1990) for a mathematical model of speculation inspired by Smith et al. (1988)).

Smith et al. (1988) also show that several replications with the same subjects were requested to attenuate the emergence of bubbles. Their results were replicated in numerous subsequent papers including King et al. (1993) who show that bubbles emerge in a variety of market environments and also conclude that three replications of the experiments were necessary to shut down irrational speculation. Dufwenberg et al. (2005) further shows that bubbles diminish even when part of the traders are experienced. The impact of experience on bubble formation is however not completely clear since more recent papers suggest that bubbles can be rekindled when experienced traders are confronted with new market parameters (Hussam et al., 2008) and when new traders enter the market in an overlapping-generation experiment (Deck et al., 2014).

Our paper echoes this experimental debate and studies the conditions under which experience can be expected to shut down irrational speculative bubbles. Our experiment based on Moinas and Pouget (2013) is not directly comparable to Smith et al. (1988)'s experiment in which traders may buy or sell as much as they want at prices and times that they choose. Our framework is nevertheless useful to study traders' adaptive learning process in speculative bubbles because it isolates the decision to speculate from other decisions (related for example to the pricing and the timing of trades). This enables to focus on a setting with a finite number of decisions to be made and a finite number of past events to be observed. At the same time, the setting is rich enough to study different types of speculative behavior: speculation due to confusion and speculation due to bets on others' confusion. Finally, the bubble game is short enough to be repeated a large number of times, thereby offering ample opportunities for subjects to learn.

Our paper also proposes a theoretical analysis of learning in speculative bubbles. It is thus related to the various theoretical papers that study speculation in the Smith et al. (1988)'s setting. Duffy and Ünver (2006) show that near-zero intelligence traders generate irrational bubbles and crashes. To establish this result, they assume that traders are endowed with some foresight ability and adopt a price setting behavior with anchoring effects and within exogenously defined bounds. Haruvy and Noussair (2006) consider a model with three types of agents, i.e., feedback, passive and speculative traders, to explain the relationship between short-sales constraints and asset prices in experimental markets. Baghestanian and Walker (2015) find that a visual anchor at the fundamental value in the first period only is sufficient to eliminate or significantly reduce bubbles in laboratory and adapt the agent-based model of Duffy and Ünver (2006) to further explore the involved mechanisms. Baghestanian et al. (2015) build on the approach of Duffy and Ünver (2006) and Haruvy and Noussair (2006) to provide a model with heterogenous agents whose parameters are estimated using the experimental data. Their model reproduces important patterns of the data and enables to classify subjects as fundamental, speculative and noise traders. Baghestanian et al. (2015) show that subjects identified as noise traders have lower cognitive skills than others. We complement these analyses by studying how learning affects speculative behavior.

Some papers have studied learning to speculate in an environment where speculation is rational. In this case, the surprising result is that speculation did not occur as much as could be expected. Duffy and Ochs (1999) for example show that subjects had difficulties coordinating on the speculative equilibrium in a money experiment. They show that this lack of speculation was related to a tendency of subjects to rely on past choices' payoffs rather than on the profit to be expected from speculation. Duffy (2001) extends the analysis of Duffy and Ochs (1999) and shows that the propensity to speculate increases when profit opportunities are more frequent or less risky in line with an agent-based model that incorporates some features of adaptive learning. We complement these articles by proposing a simple irrational bubble experiment in which a more general adaptive learning model can be tested and estimated.

In our bubble game, speculation may partly be due to failures of backward induction, and our research studies the consequences of failures of backward induction on speculative behavior, an important phenomenon in financial markets. Failures of backward induction have been identified in the experimental economics literature (see, e.g., Embrey et al., 2018 for finitely repeated prisoner's dilemma, and Andreoni (1995) for finitely repeated public good games; see also Binmore et al. (2002)) and, in particular, in the context of the centipede game (see, e.g., McKelvey and Palfrey, 1992; Nagel, 1995; Ponti, 2000; Johnson et al., 2002). Learning in this context has also been studied by experimentalists. For example, Nagel and Tang (1998) propose an experiment to study learning in the normal-form centipede game. Their results suggest that learning is not necessarily conducive to equilibrium. Duffy and Nagel (1997) show that directional learning in a beauty contest game can induce some agents to adjust their play away from equilibrium as they get experienced. This phenomenon can also be present in the adaptive learning model that we study and we show that it plays an important role in the fact that experience might induce more bubbles in the short-run.

Empiricists have also found clever ways at identifying learning and its impact on various financial and macroeconomic variables. For example, Kaustia and Knüpfer (2008) find that the outcomes of IPOs in which they have participated affect future investment behavior of individuals. Malmendier and Nagel (2011, 2016) show that investors' expectations regarding stock returns and inflation depend on their own economic experience rather than on the entire sample of information available. This indicates that learning based on experience is at work in the field. We complement these studies by showing how adaptive learning can fuel speculative bubbles. In particular, our theoretical analysis based on adaptive learning complements the analyses of Caginalp and Ermentrout (1990), Caginalp and DeSantis (2011), and Barberis et al. (2015) which show that extrapolative behavior can trigger bubble formation.

3. The bubble game and the experimental setup

We consider the bubble game designed by Moinas and Pouget (2013).

Asset. In this game, there is one asset that generates no cash flow and this is common knowledge to all market participants. This is a desirable feature that enables one to determine the fundamental value without relying on assumptions regarding the risk preferences of market participants. This fundamental value is zero.

Trading. The financial asset can be traded in a sequential market. We focus on the case in which there are three traders in each market. Traders are ex-ante equally likely to be in each position in the market sequence, that is, any trader can be first, second, or third with probability $\frac{1}{3}$. Traders are not told their positions in the sequence.

Prices. Prices are assumed to be exogenous and are powers of 10. The first trader, i.e., the trader in the first position, is offered an opportunity to buy the asset at price $P_1 = 10^n$, where n is random and follows a geometric distribution of parameter $\frac{1}{2}$, truncated at $P = 10^K$:

$$\mathbb{P}(n = k) = \begin{cases} \frac{1}{2}^{k+1} & \text{if } k \in \{0, 1, 2, \ldots, K-1\}, \\ \frac{1}{2}^{k} & \text{if } k = K. \end{cases} \tag{1}$$

In this setup, 10^K is the largest potential price that can be offered to the first trader.

If a trader decides to buy, he proposes to resell at a price that is 10 times higher than the price at which he bought: if a trader i decides to buy the asset at price P_i, where $i = 1, 2, 3$, he proposes the asset to the next potential trader at a price $10P_i$. That is, each subsequent trader is (potentially) offered an opportunity to buy at a price that is ten times higher than the previous price. With three traders, the cap 10^K on the first price thus translates into a cap of 10^{K+2} on the highest potential price in the bubble game.

Trading profits. At the beginning of the game, each trader is endowed with a capital of 1 unit of the Experimental Currency Unit (ECU) that he can use to buy the asset. If a trader in a previous position, if any, decided not to buy, a trader keeps his capital of 1 ECU and obtains a trading profit of 0 since the game has stopped before his decision could matter. If he is proposed to buy at a price P_i and chooses not to buy the asset, a trader's profit is also 0.

If the trader decides to buy the asset at price P_i, additional capital is required if $P_i > 1$. This additional capital (that is, $P_i - 1$) is provided by an outside financier. The experimenter plays the role of the outside financier for all players. Payoffs are then divided between the trader and the financier in proportion to the capital initially invested: a fraction $1/P_i$ for the trader and a fraction $(P_i - 1)/P_i$ for the financier. First, if the trader is unable to resell the asset to the next trader, both the subject and the outside financier end up with a payoff of 0 and loose the amount invested since the asset generates no

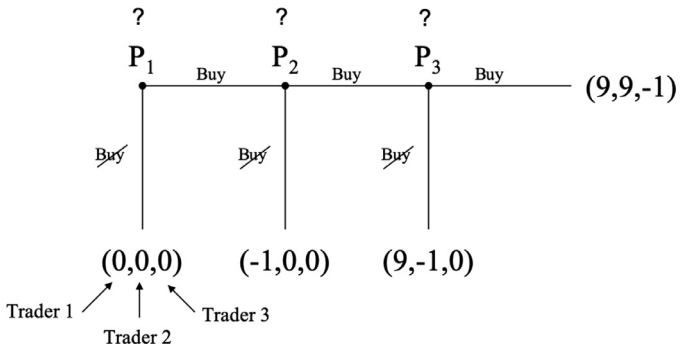


Fig. 1. Timing and profits of the bubble game. The bubble game features a sequential market. Traders can be first, second or third in a market sequence. Question marks emphasize the fact that traders are ex-ante equally likely to be first, second or third, and to be offered to buy at prices P_1, P_2, or P_3, respectively. This figure displays traders' trading profits. Trading payoffs equal the profit plus one corresponding to the initial endowment of one unit capital, that traders invest if they decide to buy the asset.

cash flow. This corresponds to a trading loss of -1 units of capital for the trader (and $-(P_i - 1)$ for the financier). Second, if the trader is able to resell at a price $10P_i$, his final wealth amounts to his share $1/P_i$ of the resale value $10P_i$, that is, 10 whatever the price P_i. His net trading profit is equal to his share $1/P_i$ of the total profit $10P_i - P_i$, that is, 9 (while the financier's trading gains amount to $9(P_i - 1)$).

Traders can thus obtain the following profits no matter the price they are proposed: 0 if the trader refuses to buy or if he is not proposed to buy, 9 if the trader buys and is able to resell, and -1 if he buys but cannot resell (either because he is last or because the next trader refuses to buy). The introduction of outside financiers allows the implementation of limit liability for traders in the experiment: the maximum potential loss of a trader is 1. The timing and trading profits of the bubble game is Fig. 1.

Informational content of prices. Although traders are not provided information on their exact positions, they can infer information on their positions from the proposed price because the cap on the first price is common knowledge to all traders. For instance, upon receiving a price of 1, a trader is sure to be first, and upon receiving a price of 10^{K+2}, a trader is sure to be last. More generally, traders can update their beliefs on their position upon observing the price they are proposed. For instance if $K = 4$ and he receives a price of 100, Bayes' rule dictates that a trader can be first, second and third with probability $\frac{1}{7}$, $\frac{2}{7}$ and $\frac{4}{7}$, respectively.[3] A cap of 1 on the first price is a specific case: upon observing the price at which they are proposed to buy, traders can perfectly infer their position in the market sequence. Observing a price of 100 reveals that one is last, a price of 10 that one is second, and a price of 1 that one is first. In order to prevent subjects from having to apply Bayes' rule, the probabilities on market positions conditional on all potential prices are provided in the experiment and are common knowledge to all market participants.

To prevent participants from discovering their position in the market sequence by hearing other subjects making choices or by measuring the time elapsed since the beginning of the game, traders play simultaneously. That is, once P_1 has been randomly determined, the first, second and third traders are simultaneously proposed prices P_1, P_2 and P_3 respectively. They are simultaneously requested to indicate whether, if they were proposed this price, they would buy or not the asset. If they decide to buy the asset, they automatically propose to resell the asset to the next potential trader. A trader's decision matters only if previous traders, if any, have all decided to buy the asset, otherwise the game has already stopped prior to his decision in the sequential market. This setup enables us to observe all three traders' decisions, even if the first trader decides not to buy (a case in which the bubble does not start and in which we could not have observed what other traders would have done). Because we do not observe traders' decisions at all potential prices but only at the price that they could be proposed given the draw of the first price, we are not in a full but in a partial strategy method context.

Backward induction and Nash equilibrium. With a cap 10^K on the first price, Moinas and Pouget (2013) show that no bubble can arise in equilibrium if rationality is common knowledge: the unique dominance solvable equilibrium involves all traders refusing to buy the asset. The reasoning is as follows. Upon receiving the maximum price, 10^{K+2}, a trader would know that he is last in the market sequence and would refuse to trade. Upon receiving the price before the last one, 10^{K+1}, a trader would anticipate that the next trader, if any, would know that he is last and would refuse to trade. As a result, he would also refuse to trade. This "infection" argument à la Morris et al. (1995) shows that, at equilibrium, all traders refuse to trade and no bubble occurs.

[3] With $K = 4$, a price of 100 can be observed in three cases: i) the trader is first and the first price drawn is 100, which occurs with probability $\frac{1}{8}$, ii) the trader is second and the first price drawn is 10, which occurs with probability $\frac{1}{4}$, and iii) the trader is third and the first price drawn is 1, which occurs with probability $\frac{1}{2}$. Therefore the probability to be first conditional on observing a price of 100 is $\frac{\frac{1}{3}\frac{1}{8}}{\frac{1}{3}\frac{1}{8}+\frac{1}{3}\frac{1}{4}+\frac{1}{3}\frac{1}{2}} = \frac{1}{7}$. Similar reasonings apply to find the probabilities to be second and third in the market sequence conditional on receiving a price of 100.

Table 1
Treatments and sessions.

Cap on the first price	No. of replications	Session	No. of subjects
1	10	A	12
1	10	D	18
1	10	F	18
1	20	H	12
10,000	10	B	18
10,000	10	C	18
10,000	10	E	18
10,000	20	G	18

Moinas and Pouget (2013) show theoretically that when traders suffer from bounded rationality (in the form of limited depth of reasoning as in cognitive hierarchy models, see, e.g., Camerer et al. (2004), or in the form of random mistakes as in the quantal response models, see, e.g., McKelvey and Palfrey (1995)), their propensity to speculate depends on the number of steps of iterated reasoning, and on the probability to be last. For a given cap 10^K on the first price, both variables depend on the price observed by the trader and on the price cap. For instance, when the cap on the first price is 1 (which corresponds to $K = 0$) so that prices perfectly reveal one's position in the market sequence, the maximum number of steps of reasoning is equal to two: zero step of reasoning is required for the last trader who receives the maximum price of 100; one step of reasoning is required for the second trader who receives a price of 10; two steps of reasoning are required for the first trader who receives a price of 1. More generally, when the cap on the first price is 10^K, the maximum number of steps of reasoning is $K + 2$.[4]

Treatments and subjects. We implement a two-by-two design. The first treatment variable is the cap on the first price: the cap is either 1 or 10,000 yielding, respectively, two and six maximum steps of iterated reasoning. The second treatment variable is the number of replications that is either 10 or 20. A total of 132 subjects have participated in our experiment. Subjects are junior and senior undergraduates in business administration at the University of Toulouse. Table 1 describes participants in the various sessions and treatments: 48 subjects have participated in an experiment with a cap at 1 and 10 replications, 54 subjects with a cap at 10,000 and 10 replications, 12 subjects with a cap at 1 and 20 replications, and 18 subjects with a cap at 10,000 and 20 replications. The treatment condition with 20 replications includes less subjects because we designed the condition for robustness purposes.

Experimental protocol. The instructions of the experiment with a cap at 10,000 are in Appendix A. Instructions include a quiz on the rules of the game, and the experiment started once all subjects were able to answer all questions correctly. The ability of all participants to understand the potential outcomes of their decisions is thus common knowledge across participants in a session.

Subjects are randomly matched across replications in a stranger design and play anonymously.[5] They are offered 1 ECU at the beginning of each replication. The exchange rate we use in the experiment is 1 Euro per ECU. Subjects are paid for all replications. After each replication, subjects are provided with information on whether they are actually proposed to trade and on their tradings gains at the current replication and over all past replications. The average, minimum and maximum gains in the experiment were, respectively, 30 Euros, 5 Euros and 134 Euros, including a 5 Euros show-up fee. The experiment lasted between one hour and one hour and a half.

4. Experimental results

In this section, we analyze the experimental data and study how learning affects bubble formation across replications.

Market outcomes. We start by analyzing behavior at the market level. Fig. 2 plots the empirical frequencies of bubbles in our experiment. We focus on the first 10 replications of the game so that we can consider all cohorts, i.e., those that played with a cap on the first price at 1 and those with a cap at 10,000. Large, medium, small and no bubbles correspond to situations in which all three subjects in a market have decided to buy the asset, only the first two subjects have bought, only the first subject has bought and no subject has bought, respectively.

The upper graph in Fig. 2 displays the results for the case in which there is a cap on the first price at 1. It shows that the market experiences less and less bubbles over time. During the first replication, the no-bubble case occurs 10% of the time on average across cohort while, during the 10th replication, it occurs 40% of the time. A Wilcoxon rank-sum test indicates that this difference is statistically significant (p-value=0.03; 20 observations corresponding to 60 subjects playing in groups

[4] When there is no cap on the first price, the bubble game allows for rational bubbles. Indeed, Moinas and Pouget (2013) theoretically show that the conditional probability to be last is constant and equal to 4/7 for all prices above 100. As a result, traders may coordinate either on a no-bubble Nash equilibrium or, they are not too risk averse, on a bubble Nash equilibrium. Because coordination complexifies the learning process, we did not investigate learning in a treatment allowing for rational bubbles. This could be interesting but is left for future investigation.

[5] The stranger design with anonymity enables to mitigate reciprocity or other-regarding motives that could induce coordination on speculation in our experiment.

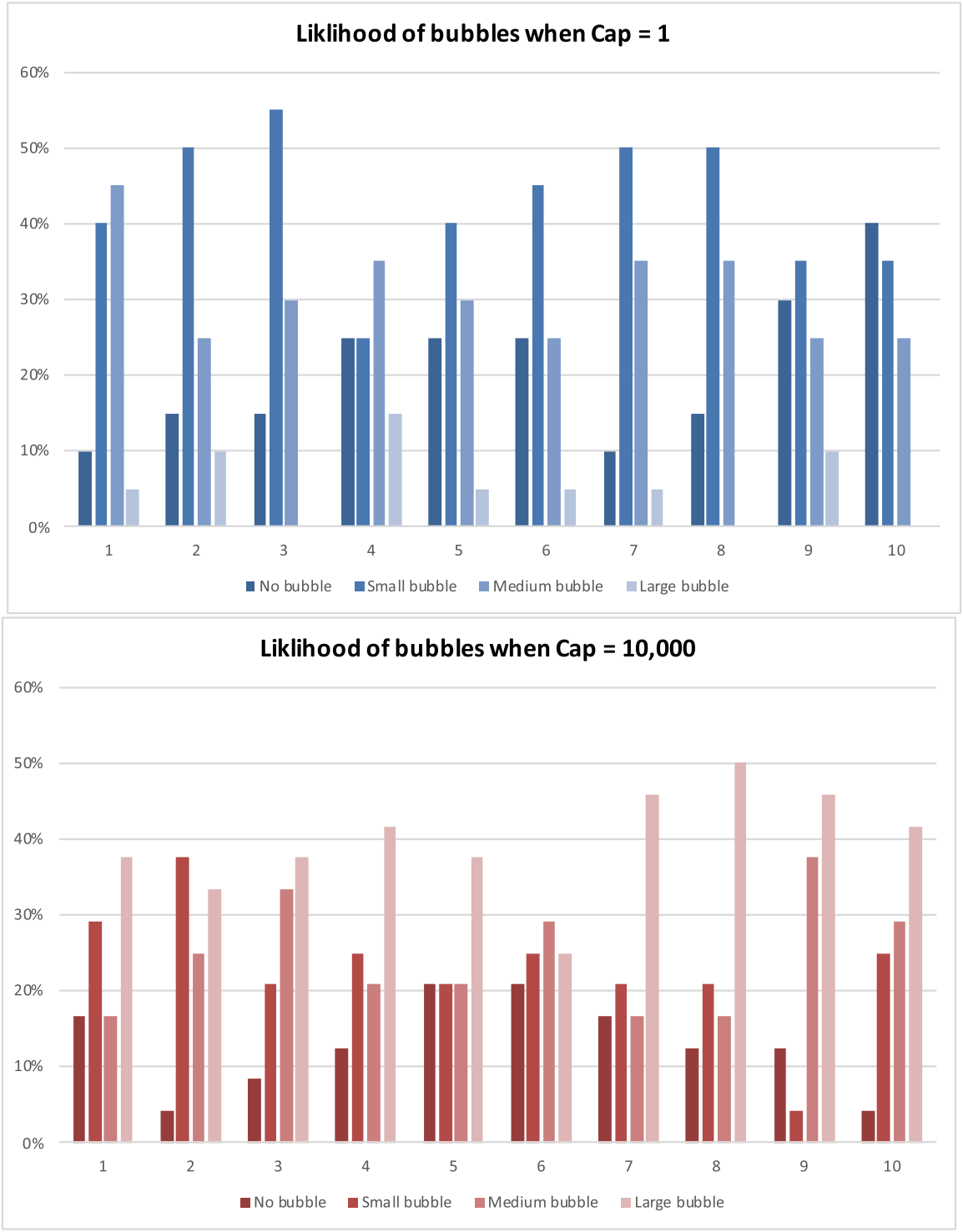


Fig. 2. Data on the frequency of bubbles. Large, medium, small and no bubbles correspond to situations in which all three subjects in a market have decided to buy the asset, only the first two subjects have bought, only the first subject has bought, and no subject has bought, respectively.

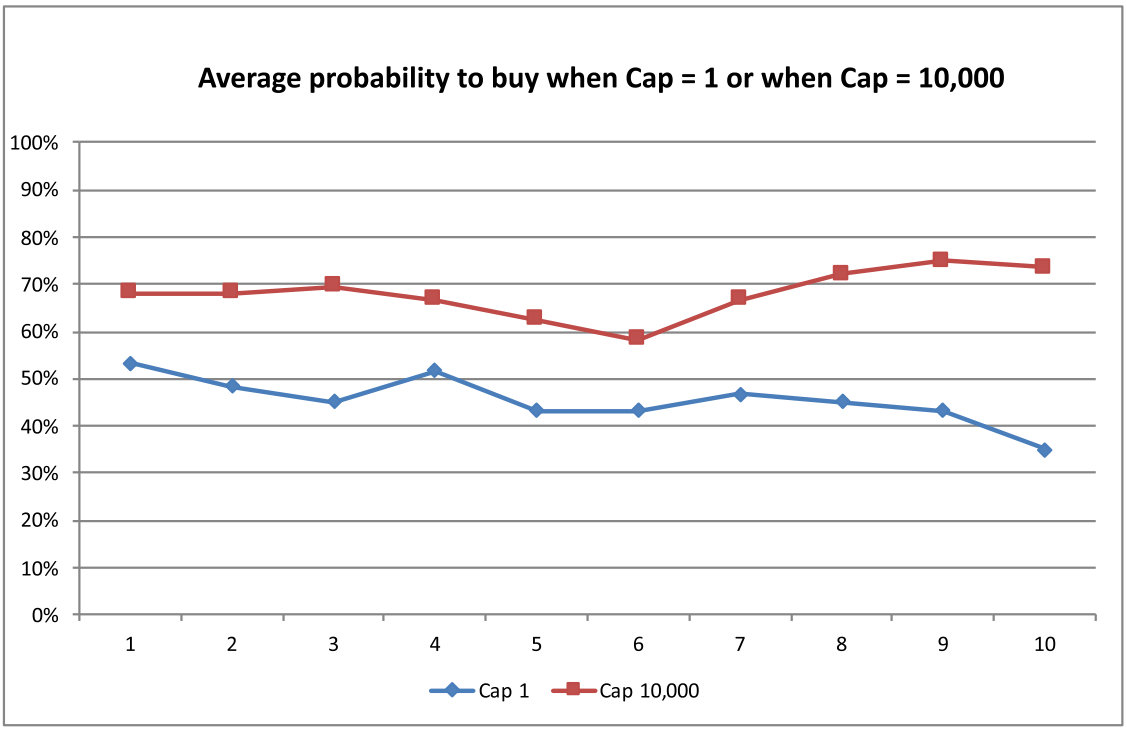


Fig. 3. Data on the average probability to speculate. Probability to buy by period averaged across subjects and prices, as measured in the data.

of three with a cap of 1).[6] As shown in the bottom graph, bubbles are not eliminated by experience when there is a large number of steps of reasoning as is the case when the cap on the first price is 10,000. In this case, during the first replication, the no-bubble case arises about 20% of the time while it occurs only less than 5% of the time during the 10th replication (p-value=0.16; 24 observations corresponding to 72 subjects playing in groups of three with a cap of 10,000). Moreover, the proportion of groups with no bubble appears significantly lower when the cap is 10,000 versus 1 at the tenth replication (p-value$<$0.01) but not at the first replication (p-value=0.53).

Fig. 2 also speaks to the average size of the bubble. The size of a bubble is set to 0 if there is no bubble, 1 if only the first trader in the market sequence speculates, 2 if only the first two traders speculate, and 3 if all traders speculate. For the case with a cap at 1, the average size of the bubble is 1.45 in the first replication (median is 1.5) and 0.85 in the tenth replication (median is 1), a difference that is statistically significant (p-value=0.03). For the case with a cap at 10,000, the average size of the bubble is 1.75 in the first replication (median is 2) and 2.08 in the tenth replication (median is 2). This difference is not statistically significant (p-value=0.33). The size of the bubbles when the cap is 10,000 appears similar to the case with a cap of 1 at the first replication (p-value=0.33). On the contrary, at the tenth replication, the size of the bubble appears significantly lower in the cap 1 case than in the cap 10,000 case (p-value$<$0.01). Overall, our experimental results suggest that learning is not likely to shut down speculative bubbles quickly when a high number of steps of reasoning is needed.[7]

Individual speculation decisions. To complement our investigation of market outcomes, we now focus on individual speculation decisions. Fig. 3 displays the evolution of the average probability to buy across replications.

When the cap on the first price is 1 (blue line), there are few steps of reasoning to reach the equilibrium and the average probability to buy shows a slowly decreasing trend. When the cap on the first price is 10,000 (red line), there is no monotonically negative trend in the probability to buy. In both cases, we still observe speculation after 10 replications. The trends observed in Fig. 3 are difficult to interpret because the average probability to buy takes into account buying decisions at different price levels and because prices observed in a given replication and in a given cohort depends on the realization of the first price. To further understand the evolution of speculative behavior in the experiment, Fig. 4 displays

[6] Because we use a stranger design, it is informative to present non-parametric tests that assume samples are independent. However, we also recognize that the samples are not perfectly independent because, even if subjects do not know the identity nor the random composition of their group, the same subjects in a given session play all the replications. The regression analyses at the individual level that we present below take into account this potential dependency by adopting a panel structure and controlling for individual fixed effects, when adequate.

[7] These results focus on the first 10 replications of experiment, irrespective of the total number of replications. For replications 11 through 20 in experiments with 20 replications, we observe the same qualitative results when the cap on the first price is 10,000. When the cap on the first price is 1, we have a low number of subjects (12) and of groups per replication (4) so it is difficult to draw strong conclusions.

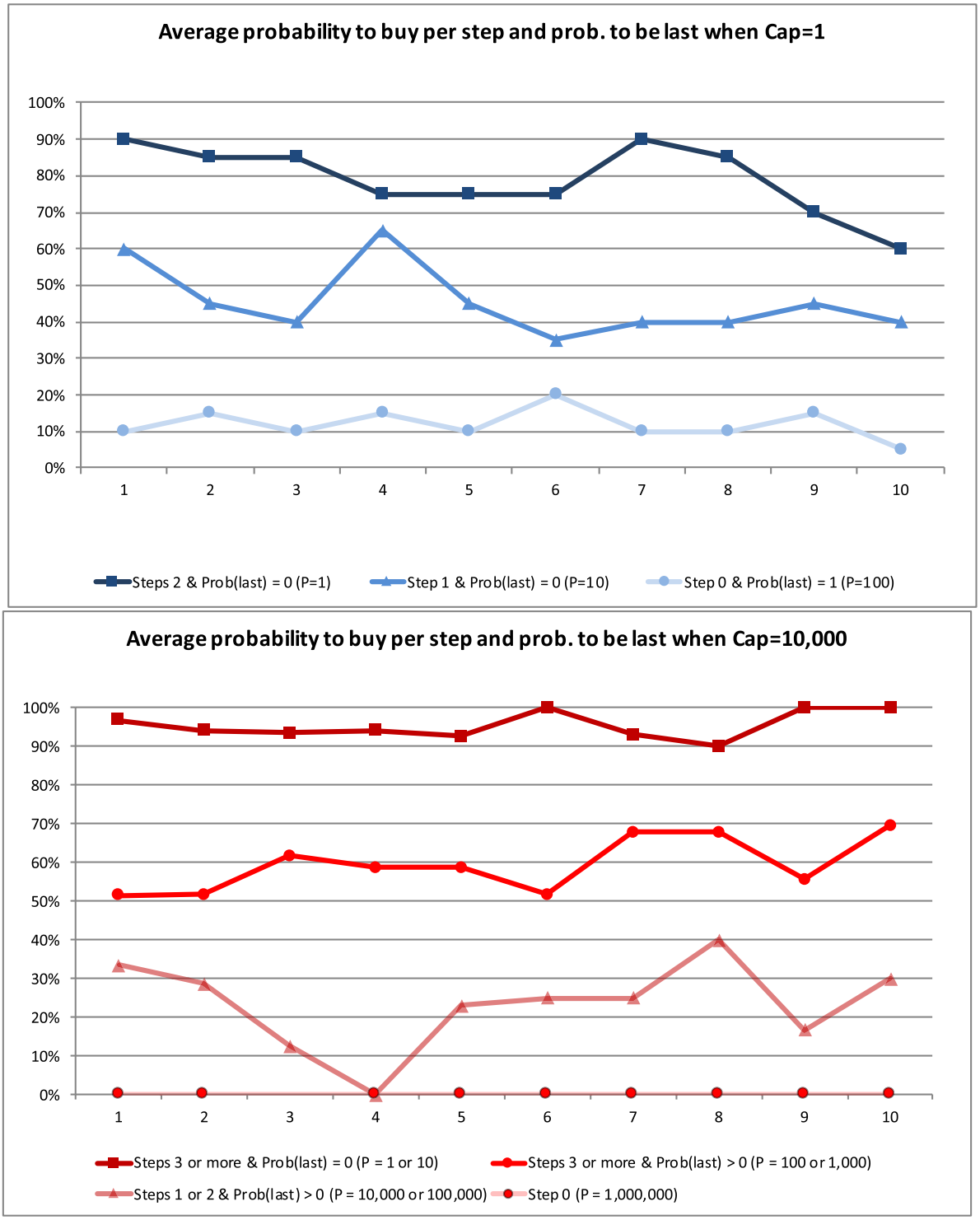


Fig. 4. Data on the probability to speculate per step and probability to be last (and price). Probability to buy per step and probability to be last by period averaged across subjects, as measured in the data.

Table 2
Probability to speculate and impact of experience.

	(1)	(2)
Constant	2.251***	4.704***
	(3.03)	(4.27)
V1: $\mathbb{1}_{Step=1 \text{ or } 2} \times \mathbb{1}_{0<P(last)<1}$	1.558	14.60
	(1.37)	(0.02)
V2: $\mathbb{1}_{Step=1 \text{ or } 2} \times \mathbb{1}_{P(last)=0}$	3.350***	4.527***
	(4.04)	(6.12)
V3: $\mathbb{1}_{Step\geq 3} \times \mathbb{1}_{0<P(last)<1}$	2.308***	16.35
	(2.83)	(0.02)
V4: $\mathbb{1}_{Step\geq 3} \times \mathbb{1}_{P(last)=0}$	5.619***	19.87
	(4.46)	(0.03)
Time		0.0934
		(0.89)
$\mathbb{1}_{Step=1 \text{ or } 2} \times$Time		0.00239
		(0.02)
$\mathbb{1}_{Step\geq 3} \times$Time		0.193*
		(1.68)
Observations	132	1230
Individual fixed effects	No	Yes

Regression of the likelihood that a subject buys the overvalued asset onto various explanatory variables that reflect the probability to be last in the market sequence, the number of steps of reasoning, and the passage of time. We control for subject fixed effects. In the panel analysis (column 2), 4 subjects were dropped because they never bought and 5 because they always bought. T-statistics are in parentheses. *, **, and *** denote statistical significance (two tailed) at the 10%, 5%, and 1% levels, respectively.

the probability to buy at different price levels across replications. The different price levels are associated with different numbers of steps of reasoning and with a different conditional probability to be last.

Fig. 4 shows that the probability to speculate in the first period is higher the lower the level of prices. This reflects the fact that lower prices tend to be associated with more steps of reasoning and lower probabilities to be last. When subjects are sure to be last, we observe no speculation at highest price when first price cap is 10,000 but a small likelihood of speculation when the first price cap is 1, indicating that some subjects may make mistakes. When subjects are sure to be first, i.e., facing a price of 1, the propensity to speculate is much larger than the one at higher prices. We also observe more speculation when the first price cap is 10,000 than that when the first price cap is 1. These results are consistent with those of the one-shot game analyzed by Moinas and Pouget (2013).

Fig. 4 also reveals that the evolution of speculative behavior depends on number of steps of iterated reasoning and on the probability to be last. When the cap on the first price is 1 (top graph), the probability to buy seems to decrease at all price levels, even if not monotonically. However, when the cap on the first price is 10,000 (bottom graph), patterns seem different across price levels. For subjects who observe a price of 1 or 10, from which they infer that they cannot be last, and that requires five or six steps of iterated reasoning, the probability to buy remains larger than 90% all along the experiment. When the price is 100 or 1,000, the probability to buy appears to increase from 50% to 70%. For the remaining very high prices, there is no clear trend (the probability to buy trends upward, downward or is flat depending on the price). Overall, there is evidence that behavior evolves with experience but not always towards less speculation.

Multivariate analysis of individual decisions to speculate. To further study the effect of experience on bubble formation and how this effect varies with the number of steps of iterated reasoning and the probability to be last, we run a series of panel logit regressions, reported in Table 2. We regress the likelihood that a subject buys the overvalued asset onto two sets of explanatory variables. First, we introduce two dummies capturing the effect of the number of steps of reasoning, namely a dummy that equals one if there are more than three steps of iterated reasoning and zero otherwise, and a dummy that equals one if there are one or two steps and zero otherwise. Two other dummies capture the effect of the probability to be last, namely a dummy that equals one if this probability is 0 and a dummy that equals one if it is strictly between 0 and 1. The four variables resulting from the interactions between these dummies represent the categories defined in Fig. 4: variable V1 captures the behavior of subjects who have received a price of 10,000 or 100,000 when the cap on the first price is 10,000; variable V2 captures the behavior of subjects who have received a price of 1 or 10 when the cap on the first price is 1; variable V3 captures the behavior of subjects who have received a price of 100 or 1000 when the cap on the first price is 10,000; variable V4 captures the behavior of subjects who have received a price of 1 or 10 when the cap on the first price is 10,000. The constant in our regression captures the behavior of subjects who observe the highest potential price and thus know that they are last.

Second, we include explanatory variables capturing the effect of experience on speculative behavior. To capture this effect, we include *Time*, corresponding to the replication number, to document whether traders speculate less over time. We expect a negative coefficient if traders learn not to speculate. However, learning may depend on the complexity of the game, as faced by traders. To capture this complexity, we interact the dummies indicating whether traders have few or

many steps of iterated reasoning, with the *Time* variable. We also control for subject fixed effects. In the analysis, 4 subjects were dropped because they never bought and 5 because they always bought.

The results of the panel logit regressions are in Table 2. The first regression specification of Table 2 replicates the analysis of the one-shot game in Moinas and Pouget (2013) on subjects' decisions in the first period of our repeated game. As in Moinas and Pouget (2013), we find that subjects are more likely to speculate and buy the overvalued asset in the hope to resell it at a profit i) when they have a chance not to be last, ii) when the probability to be last decreases (keeping constant the number of steps of iterated reasoning, subjects are more likely to buy when the probability to be last is equal to zero than when it is strictly between 0 and 1), and iii) when the number of steps of reasoning increases (keeping constant the probability to be last, subjects are more likely to buy when there are 3 or more steps of reasoning than when there are 1 or 2 steps).

In the second regression specification of Table 2, we introduce the time variable to test whether experience affects individual behavior. We find that the propensity to speculate decreases over time when subjects have few steps of iterated reasoning (this result being statistically insignificant), but increases when subjects have more than three steps of iterated reasoning (this result being significant at the 10% level). Although the coefficients are not or only marginally significant significant, this result indicate that speculative behavior does not always decline with experience, especially when the game requires a high number of steps of reasoning. In this case, speculative behavior seems on the contrary to be reinforced with experience.

5. A theory of bubbles based on adaptive learning

Our empirical analyses imply that learning does not always reduce speculation, depending on the subjects' probability to be last and the number of steps of reasoning. However, the underlying mechanisms through which learning induces/reduces speculative bubbles are not clear. Different from Smith et al. (1988), our simple bubble game with finite number of decisions and past events enables us to apply and test adaptive learning models. In this section, we present and analyze a model based on adaptive learning to understand traders' learning process. Then we estimate the learning model on our experimental data to identify the existence of learning and the type of learning at work.

5.1. The adaptive learning model

We capture the adaptive behavior of traders using a simplified version of Camerer and Ho (1999)'s Experience-Weighted Attraction (EWA) model.[8] In the EWA learning model, an agent's choice of action in a set of potential actions is probabilistic. The probability to choose a given action depends on its "attraction" and the attractions of other actions available according to some functional form (e.g., logit). These actions' attractions reflect initial predispositions and are then updated based on actions' past performance. This enables the model to capture the evolution of play with experience. In this section, we will first introduce how attractions are updated based on past performance, and then explain how the attractions are transformed into choice probabilities for traders.

Updating attractions. In our game, the action space for traders is $\{B, \varnothing\}$, where B stands for a decision to buy, $\varnothing$ for a decision not to buy. Let us denote by $A_i^j(t|P)$ trader i's attraction of action $j \in \{B, \varnothing\}$ conditional on price P at period t. In Camerer and Ho (1999), trader i updates the attraction for the action $a_i(t)$ that he actually plays at date t from period t to period $t+1$ according to the profit $\pi_i(a_i(t), a_{-i}(t))$ he received at date t (that also depends on the actions chosen by the other traders, $a_{-i}(t)$). This captures reinforcement learning. The EWA model also nests belief-based learning: traders may update the attraction for an action that they have not played, imagining the profit they would have received from this action given the actions played by other traders.

Adapting the EWA model to our repeated trading game, the attraction of action j for trader i at a given price P at period $t+1$ is expressed as follows:

$$A_i^j(t+1|P) = \begin{cases} A_i^j(t|P) + \left[\delta + (1-\delta)\mathbb{1}_{j=a_i(t)}\right]\pi\left(j, a_{-i}(t)\right) & \text{if condition } \mathbb{A} \text{ is true,} \\ A_i^j(t|P) & \text{otherwise,} \end{cases} \tag{2}$$

where condition $\mathbb{A}$ refers to the case where i is proposed price $P < 10^{K+2}$ and he chooses to buy, or i is proposed the highest price $P = 10^{K+2}$. $\pi\left(j, a_{-i}(t)\right)$ is the profit received by trader i if he chooses action j given that the next trader in the market sequence chooses action $a_{-i}(t)$.[9] $\mathbb{1}_{j=a_i(t)}$ is an indicator function, which is equal to 1 if $j = a_i(t)$, i.e., if action j has been chosen by agent i at date t, and 0 otherwise. Parameter δ is the imagination parameter that controls how much agents are able to display counterfactual reasoning. When $\delta = 0$, trader i updates the attraction of action j only if it is selected at period t, while when $\delta > 0$, trader i updates the attraction of action j no matter whether it is actually chosen or not, provided that he is proposed the price P.

[8] Our version of the EWA learning model corresponds to the original version of Camerer and Ho (1999) with $\rho = 0$ and $\phi = 1$.

[9] We calibrate the model using trading profits, i.e., $\pi(B, \varnothing) = -1$, $\pi(B, B) = 9$ and $\pi(\varnothing, .) = 0$. Simulations using instead traders' payoffs generate similar results.

The EWA learning model intends to capture both the law of actual effect and the law of simulated effect. The law of actual effect means that the attraction of an action is adjusted only if this action has been selected ($\delta = 0$). If the action generates a positive profit, this action will be more attractive, otherwise, it will be less attractive. This law is at the core of reinforcement learning (see, for example, Roth and Erev, 1995).

The law of simulated effect indicates that the attraction of an action is adjusted according to the profit it could have generated even if it has not been selected ($\delta > 0$). This law is at the core of belief-based learning (see, for example, Fudenberg and Levine, 1998). In our framework, condition $\mathbb{A}$ ensures that the agent can apply counterfactual reasoning to determine how much he would get if he played a different action at the proposed price.[10] In other cases, the attractions are not updated either because counterfactual reasoning cannot be applied when i is proposed price $P < 10^{K+2}$ and he chooses not to buy, or price P is not proposed.

To close the system, the initial values of $A_i^j(0|P)$ need to be specified. Following Camerer and Ho (1999), we set $A_i^j(0|P) = 0$, i.e., the initial attractions for all actions are set to be 0.

Transforming attractions into probabilities. Actions' Attractions determine their probabilities to be chosen. Let $Pr_i^j(t|P)$ be the probability with which trader i chooses action j conditional on price P. The attractions are transformed into choice probabilities through a logit function as follows:

$$Pr_i^j(t+1|P) = \frac{e^{\lambda A_i^j(t|P)}}{e^{\lambda A_i^j(t|P)} + e^{\lambda A_i^{-j}(t|P)}}, \tag{3}$$

where $-j$ is the other action of trader i and λ represents the responsiveness of traders to attractions. Eq. (3) indicates that the probability for trader i to choose action j is determined by the attraction of action j relative to that of the other potential action.

5.2. Analytical results

We study the adaptive learning model and the mechanisms underlying the impact of learning on speculative bubbles. We focus on the long run and the short run. The intermediate case is much more difficult to analyze but it can be understood qualitatively, by comparing the short and the long run, or quantitatively, thanks to simulations – as developed in Section 5.3. The proofs are in Appendix B.

Our first proposition focuses on the long run convergence of our adaptive learning model applied to the bubble game. This analysis is an application of the more general case studied in Milgrom and Roberts (1991).

Proposition 1. *Assume that $\lambda > 0$. In the long run, the trading game converges to the no bubble equilibrium.*

This result comes from the fact that the bubble game with a price cap is dominance solvable. The trader with the highest price looses money by speculating. Adaptive learning will thus lead to a decrease in the probability to buy the asset at this price towards zero. Once this probability is low enough, the trader with the second highest price will also see his probability to buy converge to zero. Eventually, all traders will see their probability to buy converge to zero.

The short run situation is very different from the long run and is summarized in the following proposition.

Proposition 2. *Assume that $\lambda > 0$. At period 1, traders randomly select between buy or no buy with equal probability. At period 2, if a trader is proposed a price of 1 or 10, inducing that he cannot be last in the market sequence, his expected probability to buy increases. If a trader is proposed a price which is strictly higher than 10 and lower than the highest price, his expected probability to buy increases if λ is small and decreases otherwise. If a trader is proposed the highest price, his expected probability to buy always decreases.*

Overall, we find that in the short run, i.e., from period 1 to period 2, the propensity to speculate may increase or decrease depending on the proposed price. A trader with the highest price would lose money by buying the asset and thus will reduce his probability to buy. However, if a trader is sure that he is not last, he is likely to gain from other traders' mistakes, since at period 1 all traders would choose to buy the asset with a high probability ($\frac{1}{2}$). Therefore, his propensity to speculate is likely to increase in period 2. Our analytical results in the short run are in line with our experimental results, as displayed in Table 2, according to which the probability to be last is an important variable that determines whether experience can reduce speculation.

Our analytical results show that learning may induce speculation in the short run but shuts down speculation in the long run. In the intermediate periods, we thus expect that a trader's propensity to speculate at a given price will decrease at some point in time as soon as the probability to speculate of traders at the next price has become sufficiently low. In the following subsection, we offer a simulation analysis to further investigate traders' speculative behavior in the intermediate term.

[10] In our experimental game, after each replication, a trader is provided information on i) whether he is offered to buy the asset, ii) if so, at what price, and iii) what is his payoff for the last replication as well as for the sum of all past replications.

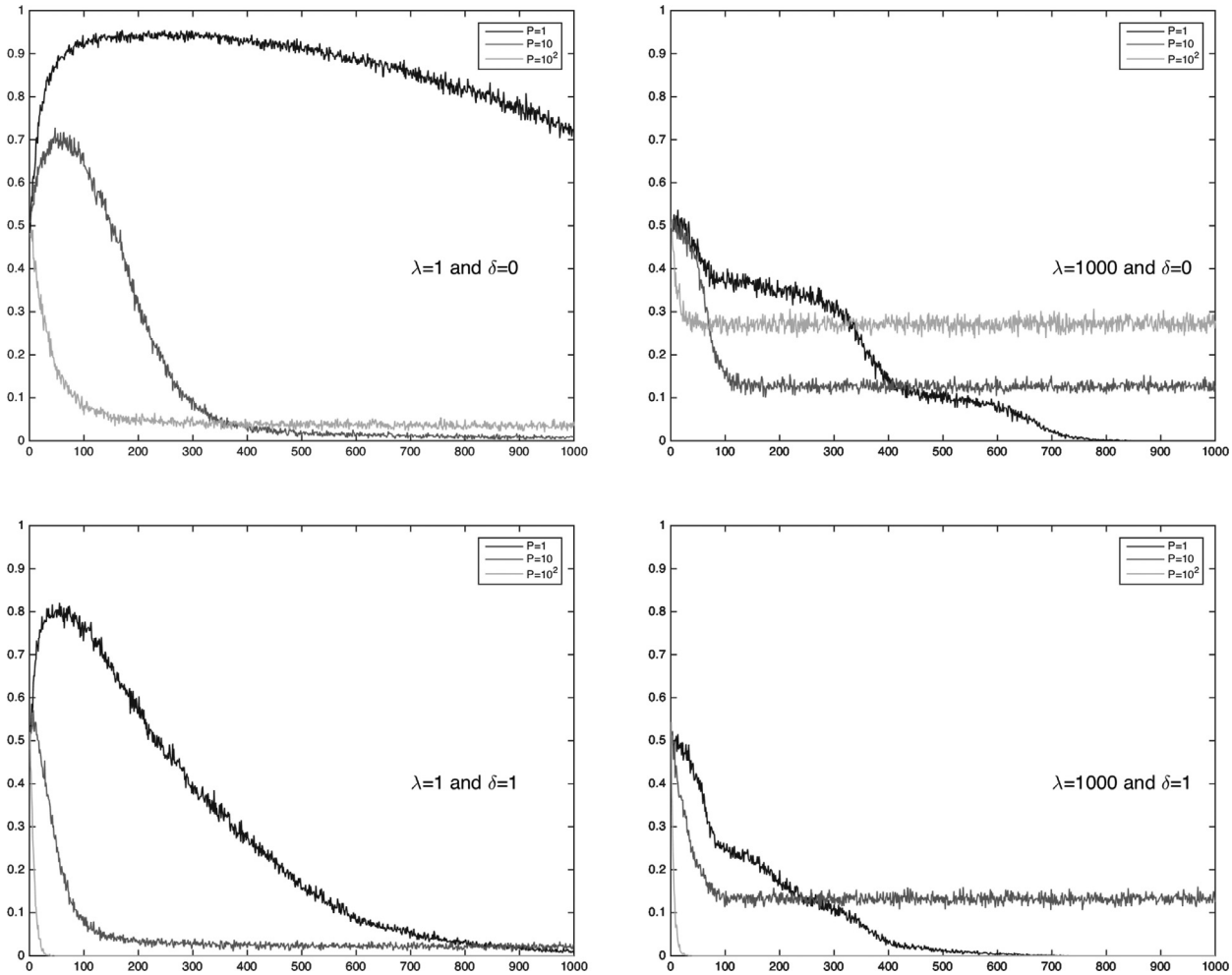


Fig. 5. Probability to speculate per price in the adaptive learning model when the cap on the first price is 1. This figure displays the average probability to buy conditional on different prices in the adaptive learning model applied to the bubble game.

5.3. Simulation results

The impact of experience on speculation during intermediate periods is difficult to analyze explicitly but it can be quantitatively understood through simulations. Our simulations consider two cases as in our experiment: price cap is 1 or 10,000. For both cases, we have a 2×2 design with δ being equal to 0 or 1, and λ being equal to 1 or 1000. Our simulation proceeds as follows: for a given set of parameters, each simulation contains 1000 independent trials; each trial contains 1000 runs, where each run represents one trading session with three artificial agents whose propensity to choose actions evolves over time.

5.3.1. Individual trading behavior: cap on the first price is 1

Fig. 5 depicts the simulated average probability to buy at different prices. We first look at the case in which $\delta = 0$ and $\lambda = 1$ (upper left graph in Fig. 5), that is, when traders only learn based on the profits of the actions that they actually choose, and are not very responsive to past gains or losses. Since the last trader can never sell back the asset, he cannot gain if he chooses to buy. Therefore, he learns not to trade the asset. By assumption, the last trader initially chooses to buy with a high probability (50% given the initial attractions). In early periods, the second trader can thus sell the asset at 10 times higher price with a high probability. He gains from trading in expectation. This initially leads to more speculation. As time goes by, the last trader learns not to speculate and the second trader is then less prone to speculate. The same pattern occurs for the first trader, the only difference being that the first trader is more inclined to buy the asset due to the initial increase in the propensity to speculate of the second trader. As a result, the convergence to no speculation is slow for the first trader.

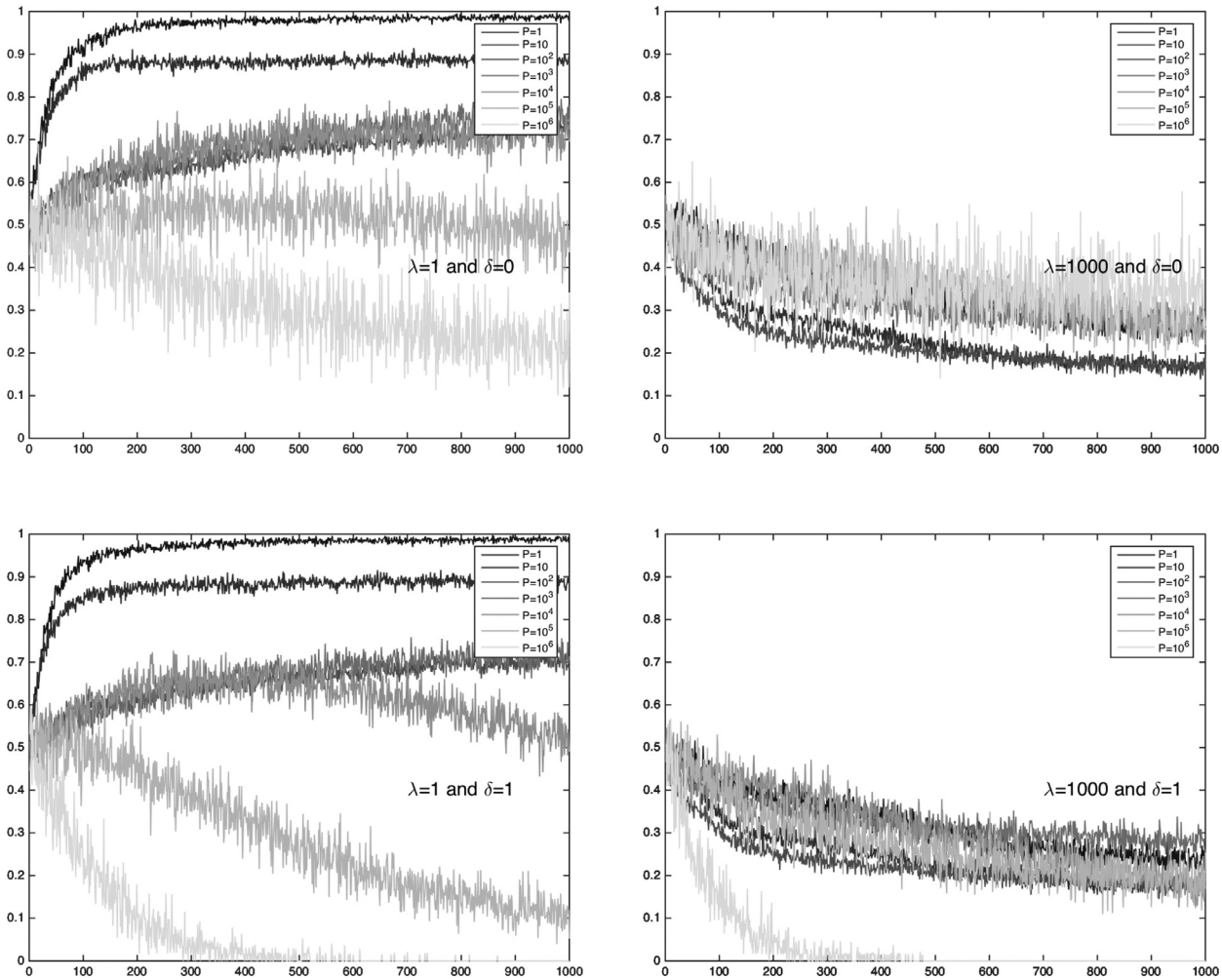


Fig. 6. Probability to speculate per price in the adaptive learning model when the cap on the first price is 10,000. This figure displays the average probability to buy conditional on different prices in the adaptive learning model applied to the bubble game.

The same forces are at play when δ increases to 1, i.e., to the maximal level of imagination. In this case, agents adapt their behavior in response to the payoffs they could have had. Our simulations (bottom left graph in Fig. 5) show that imagination leads to quicker learning. Initially, agents who are proposed the maximum price learn quickly not to speculate, which increases the speed at which the second trader learns not to buy, which in turn induces the occurrence of bubbles to decrease more rapidly than in the case without imagination.

An increase in traders' responsiveness to attractions, λ, reduces (but does not eliminate) speculation when $\delta = 0$ (upper right graph in Fig. 5): the probability to buy of the last trader converges even more slowly to zero. Indeed, when traders early in the market sequence have already learnt not to speculate, the last trader, who has no imagination when $\delta = 0$, is very rarely offered the opportunity to learn because he has few chances to be offered the opportunity to buy.

These results on individual behavior shed some light on the aggregate market behavior. The fact that the first traders in the market sequence initially learn to speculate indicates that the likelihood of bubbles initially increases. It thus takes some time for this likelihood to converge to zero but we know it will do so given that traders who are last in the market sequence learn not to speculate. Second, imagination induces the first traders in the market sequence to learn more strongly to speculate in the short run but also to learn more quickly to converge to no speculation. This suggests that in the presence of imaginative traders, markets would experience stronger but shorter speculative episodes.

5.3.2. Individual trading behavior: cap on the first price is 10,000

As shown in Fig. 6, raising the cap on the first price to 10,000 significantly fosters traders' speculation at all price levels. Traders' propensity to speculate is very high even after 1000 runs: the probability to buy the asset at prices of 1, 10, 100 and 1000 ends up greater than 70% when $\lambda = 1$. When $\lambda = 1000$, the probability to buy decreases with experience but at a slower pace than when the price cap is 1.

Table 3
Probability to speculate and adaptive learning.

	(1)	(2)	(3)
Constant	5.692***	6.480***	7.628***
	(5.70)	(6.14)	(5.40)
V1: $\mathbb{1}_{Step=1 \text{ or } 2} \times \mathbb{1}_{0<P(last)<1}$	14.60	14.44	14.67
	(0.02)	(0.02)	(0.03)
V2: $\mathbb{1}_{Step=1 \text{ or } 2} \times \mathbb{1}_{P(last)=0}$	4.748***	5.056***	6.241***
	(8.55)	(8.49)	(5.68)
V3: $\mathbb{1}_{Step\geq 3} \times \mathbb{1}_{0<P(last)<1}$	17.46	17.31	16.50
	(0.02)	(0.02)	(0.04)
V4: $\mathbb{1}_{Step\geq 3} \times \mathbb{1}_{P(last)=0}$	21.06	20.91	20.14
	(0.03)	(0.03)	(0.05)
Time			0.125
			(0.92)
$\mathbb{1}_{Step=1 \text{ or } 2} \times$Time			0.113
			(0.77)
$\mathbb{1}_{Step\geq 3} \times$Time			0.0768
			(0.52)
Accumulated gains	0.00377	0.00367	0.00461
	(0.44)	(0.42)	(0.53)
Max price $\times \mathbb{1}_{P(last)<1}$	0.159		
	(0.47)		
Max bought $\times \mathbb{1}_{\Pr(last)<1}$		2.317***	2.663***
		(3.45)	(3.27)
Max not bought $\times \mathbb{1}_{\Pr(last)<1}$		0.671*	0.738*
		(1.74)	(1.73)
Bought lost $\times \mathbb{1}_{P(last)<1}$	1.000***	1.090***	1.414***
	(3.80)	(3.91)	(4.27)
Bought won $\times \mathbb{1}_{P(last)<1}$	0.690**	0.572**	0.109
	(2.47)	(1.99)	(0.32)
Observations	1230	1230	1230
Individual fixed effects	Yes	Yes	Yes

Regression of the likelihood that a subject buys the overvalued asset onto various variables that reflect the profits or losses of past actions. We control for subject fixed effects. In the panel analysis, 4 subjects were dropped because they never bought and 5 because they always bought. T-statistics are in parentheses. *, **, and *** denote statistical significance (two tailed) at the 10%, 5%, and 1% levels, respectively.

Our theoretical investigations are in line with the experimental results according to which a trader's propensity to speculate depends on many factors, such as his probability to be last or the number of steps of iterated reasoning (captured by the price cap). Our simulations indicate that convergence to the no bubble equilibrium is slow, which is consistent with our results that 10 or even 20 replications of the simple bubble game are not enough to shut down speculation.

5.4. Testing the adaptive learning model

Our adaptive learning model provides insights on the dynamics of learning and the underlying forces behind the evolution of bubbles: our investigations show that payoff responsiveness and imagination may be important in shaping learning in speculative bubbles. In this section, we go one step further to test in our data for the presence of adaptive learning as modelled above and identify the type of adaptive learning at work, i.e., whether reinforcement learning or belief-based learning is dominant, in our experimental data. We use both regression and structural estimations.

5.4.1. Regression estimation

To start collecting evidence on the factors important in adaptive learning, we run regression analyses that build on the one presented in Table 2. We include additional explanatory variables that are related to adaptive learning and measure whether past outcomes influence individual speculation. The variable *Accumulated gains* is the sum of the profits earned by the subject in the previous replications. The dummy variable *Bought Lost* equals 1 if in at least one previous replication, the subject has bought the asset but could not resell it, and 0 otherwise. The variable *Bought Won* equals 1 if in at least one of the previous replications, the subject has bought the asset and was able to successfully sell it, and 0 otherwise. The dummy *Max price* is equal to one if the subject has already observed the maximum price, that is $P = 100$ when the cap on the first price is 1 or $P = 1,000,000$ when the cap is 10,000, in at least one of the previous replications. All these variables are interacted with a dummy $\mathbb{1}_{P(last)<1}$ that takes value one if the trader is not sure to be last in the current replication. Column (1) of Table 3 shows that, the propensity to speculate goes down after a subject has bought the asset but was unable to resell it, and goes up after a subject has experienced a profit from speculating in the past (the statistical significance of this second result is weaker: in the third column of Table 3 in which we control for time trends, the sign of the coefficient is not affected but the statistical significance disappears). The fact that they have already been proposed the highest potential price

seems not to affect the propensity to speculate, indicating that their speculation is not driven by their lack of understanding of the game structure.

We now study whether reinforcement or belief-based learning is at work in the experiment. As explained in Section 5.1, a trader can use his imagination to compute the profit he would have received if he had taken the other action only in cases in which he was proposed to purchase the asset and actually chose to buy and in which he was proposed to buy at highest possible price and chose not to buy. However, since the profit from not buying is equal to zero, imagination would not affect the attraction of the action not to buy. Therefore, imagination, measured by δ, only matters when a trader receives the highest possible price and decides not to buy. In this case, the trader can imagine he would have lost -1 if he had bought the asset. This should lead him to decrease his propensity to buy at the highest possible price. This offers us a unique opportunity to distinguish reinforcement learning and belief-based learning by testing whether subjects adjust their propensity to speculate differently if they have bought or not at the maximum price in previous replications.

In the regressions, we decompose the dummy *Max price* in two, depending on the decision of the trader when he is proposed the maximum price: he may either have decided to buy, dummy *Max bought* is equal to 1 if the subject has observed the maximum price and bought the asset in at least one previous replication, or not, *Max not bought* equals 1 if the subject has observed the maximum price and but has not bought in at least one replication. A negative coefficient for the former would be consistent with reinforcement learning, while a negative coefficient for the latter would suggest that imagination matters. The two variables also interact with dummy $\mathbb{1}_{P(last)<1}$. Columns (2) and (3) of Table 3 show that subjects who had bought (and lost) when they had observed the maximum price tend to buy significantly less, an evidence rather in favor of reinforcement learning. However, we find no evidence for belief-based learning since no significant effect was found on the probability to speculate for subjects who had not bought when offered the maximum price. These regressions focus on the determinants of the probability to buy when the subjects is not offered the maximum price, i.e., $\mathbb{1}_{P(last)<1}$. Next subsection offers a more suited test for the presence of adaptive learning.

5.4.2. Structural estimation

We estimate our adaptive learning model on the data generated in our experiment using the Maximum Likelihood technique. This is useful in order to identify learning and determine the type of adaptive learning that is at work in our data: by estimating the adaptive learning model of our bubble game, we can determine whether people tend to use the law of actual effect only (i.e., whether the estimate of the imagination parameter δ is zero) or whether they also use the law of simulated effect (the estimate of δ is larger than zero). Moreover, such estimation is useful to evaluate the strength of traders' imagination in a bubble episode.

Initial attractions are estimated from the behavior in the first replication of the experiment and in the one-period experiment of Moinas and Pouget (2013).[11] For the Maximum Likelihood estimation, we restrict the set of parameter values as follows: δ is between 0 and 1 with increments of 0.05, and λ between 0 and 10 with increments of 0.05. We compute the likelihood of the model for all the potential parameters' values. The estimated parameters correspond to the ones associated with the highest likelihood.

To compute confidence intervals for the parameters, we adopt a bootstrapping approach: we estimate the model several times after dropping one or more replications at the end of the experiment or after dropping one session. We do this in order to keep the time dependency that is inherent to repeated experiments. We thus have 8 estimations after dropping one of the eight sessions, 1 estimation after dropping the last period, 8 estimations after dropping one of the eight sessions and the last period, 1 estimation after dropping the last two periods, 8 estimations after dropping one of the eight sessions and the last two periods, 1 estimation after dropping the last three periods, 8 estimations after dropping one of the eight sessions and the last three periods. We build 89% confidence intervals by dropping the two lowest and the two largest estimates of each parameter out of the 35 estimations.

The results are displayed in Table 4. The first column corresponds to the model estimation. As we explained before, the parameter δ can be only be identified for subjects who are proposed the highest price and decide not to buy. The three other columns are offered for comparison. The second column corresponds to the case in which δ is set to 0, i.e., traders learn but have no imagination; the third column corresponds to the case in which δ is set to 1, i.e., traders learn and have full imagination. Comparing these two columns enables one to assess the importance of the law of simulated effect in our bubble experiment. Finally, the last column provides the likelihood of a model in which there is no learning, i.e., agents' behavior is fixed and similar to their behavior in the first period.

The results show that in our data, the learning model has a higher likelihood than the fixed behavior model. The difference in log-likelihood is statistically significant according to a Vuong test (p-value$<$0.01). This is in line with our previous regression results suggesting that adaptive learning is at play in our experiments and is an important element in speculative behavior.

We also find that imagination does not play a large role in our experiment. According to Vuong tests, the maximum likelihood of the learning model is significantly larger when δ is unconstrained than when it is constrained to 0 or to 1 (p-values=0.03). The improvements however appear relatively small compared to the improvement in maximum likelihood

[11] We used data from the one-shot bubble game of Moinas and Pouget (2013) in order to have observations at all prices, which is not the case in the present experiment.

Table 4
Maximum likelihood estimations of the imagination and responsiveness parameters of the adaptive learning model.

	Unconstrained δ	$\delta = 0$	$\delta = 1$	No learning
λ	0.20	0.20	0.20	> 0
δ	0.45	0.00	1.00	
Max L	571.17	571.33	571.36	594.73
Contribution of cap $= 1$				
Max L	275.92	276.08	276.11	290.80
Contribution of cap $= 10,000$				
Max L	295.25	295.25	295.25	303.93
Bootstrapped confidence intervals				
min λ	0.15	0.15	0.15	> 0
max λ	0.25	0.25	0.20	
min δ	0.00			
max δ	0.50			
Average Max L	425.96	426.01	426.30	438.90

This table shows the results of the Maximum Likelihood Estimation of the imagination parameter δ, and responsiveness parameter, λ, of our adaptive learning model of the bubble game. Initial attractions are set up to fit the behavior of subjects as displayed in the first period and in Moinas and Pouget (2013). Confidence intervals are obtained by bootstrapping. 35 estimations are done taking into account the time dependence: 8 estimations after dropping one of the 8 sessions, 1 estimation after dropping the last period; 8 estimations after dropping one of the 8 sessions and the last period; 1 estimation after dropping the last two periods; 8 estimations after dropping one of the 8 sessions and the last two periods; 1 estimation after dropping the last three periods; 8 estimations after dropping one of the 8 sessions and the last three periods. 89% confidence intervals are obtained by dropping the 4 most extreme estimates of each parameter out of the 35 estimations.

between the learning models and the no-learning benchmark. This is not to say that imagination is not important for speculation. Indeed, in our experiment, when a subject decided not to buy, he was not told what the next subject in the market sequence (if any) wanted to do. He thus had no data on which to apply his potential imagination. To study this issue further and test whether imagination is important for speculation decisions, it would be interesting to run an experiment in which subjects were told what the other subjects wanted to do.

6. Conclusion

In this paper, we study whether traders' experience reduces their propensity to speculate. We theoretically study a financial market populated by adaptive traders. Following Camerer and Ho (1999)'s Experience-Weighted Attraction (EWA) learning model, these traders are assumed to adjust their behavior according to actions' past performance. The EWA model nests reinforcement learning, on the one hand, and belief-based learning, according to which agents use their imagination to do counter-factual reasoning, on the other hand.

We focus on the Bubble Game designed by Moinas and Pouget (2013) in which agents sequentially trade a worthless asset and agents do not always know where they stand in the market sequence. In the version of the Bubble Game we consider, because there is a cap on the maximum price that can be achieved, no rational bubbles can form.

The learning model shows that, in the long-run, the market converges to the unique no bubble equilibrium. However, we also find that learning may initially increase traders' propensity to speculate: this is because as long as not all traders have learned not to speculate, some traders can make substantial profit by speculating. In the short run, more experienced traders thus create more bubbles. Moreover, we show that this effect is stronger when the price cap is higher so that there are more steps of reasoning to reach equilibrium.

We provide experimental evidence that is consistent with these theoretical results. First, adaptive learning is at work in speculative markets: subjects' past payoffs have a statistically significant impact on their propensity to speculate. Secondly, the propensity to speculate and the occurrence of bubbles are reduced with experience when the price cap is low but not when it is high. Our further tests identify that reinforcement learning rather than belief-based learning is the main mechanism at work in our experiment.

In future work, it would be interesting to study how traders learn when they can observe other traders' behavior. This would allow one to investigate how observing a bubble inflating affects the propensity to speculate of a trader who has not yet decided to ride the bubble. In the Bubble Game, this could be captured by enabling agents to observe what the next trader in the market sequence has done.

Appendix A. Instructions

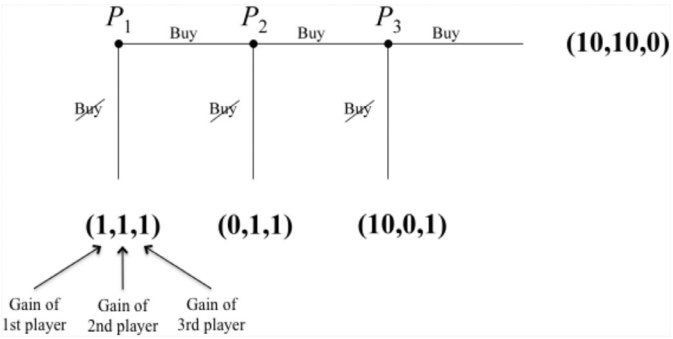
Welcome to this market game. Please read carefully the following instructions. They are identical for all participants. Please do not communicate with the other participants. The game will last approximately one hour, including the reading of the instructions.

You will be able to earn some money during the game. Your total gain will be paid to you on March 27th, 2017.

Trading process

To play this game, we form groups of three players. Each player is endowed with one experimental currency unit (ECU) that can be used to buy an asset. Your task during the game is to decide whether you want to buy or not the asset. This asset does not generate any dividend. If the asset price exceeds 1 ECU, you can still buy the asset. We indeed consider that a financial partner (who is not part of the game) provides you with the additional capital and shares profits with you according to the respective capital invested. The market proceeds sequentially. A proposal to buy at a price $P1$ is made to the first player. If he buys, he proposes to sell the asset to the second player at a price which is ten times higher, $P2 = 10 * P1$. If the second player accepts to buy, the first player ends up the game with 10 ECU. Otherwise, the first player ends up with nothing. If he or she accepted to buy, the second player then proposes to sell the asset to the third trader at a price $P3 = 10 * P2 = 100 * P1$. If the third player buys the asset, the second player ends up the game with 10 ECU. Otherwise, the second player ends up with nothing. The third player does not find anybody to whom he or she can sell the asset. Since this asset does not generate any dividend, he or she ends up the game with nothing. If a player refuses to buy, he or she ends up the game with 1 ECU (so do all the player after him or her in the market sequence).

At the beginning of the game, players do not know their position in the market sequence. Positions are randomly determined with one chance out of three for each player to be first, second or third. The game is summarized in the following figure.



Proposed prices

The price $P1$ that is proposed to the first player is random. This price is a power of 10 and is determined as follows:

Price $P1$	Probability that this price is realized
1	1/2 (50%)
10	1/4 (25%)
100	1/8 (12.5%)
1,000	1/16 (6.25%)
10,000	1/16 (6.25%)

Players' decisions are made simultaneously and privately. For example, if the first price $P1 = 1$ is drawn, the prices that are simultaneously proposed to the three players are: $P1 = 1$ for the first player, $P2 = 10$ for the second player, and $P3 = 100$ for the third player. Identically, if the first price $P1 = 10,000$ has been drawn, the prices that are simultaneously proposed to the three players are: $P1 = 10,000$ for the first player, $P2 = 100,000$ for the second player, and $P3 = 1,000,000$ for the third player.

The price that you are been proposed can give you the following information regarding your position in the market sequence:

- if you are proposed to buy at a price of 1, you are sure to be first in the market sequence;
- if you are proposed to buy at a price of 10, you have one chance out of three to be first and two chances out of three to be second in the market sequence. You are sure not to be last;
- if you are proposed to buy at a price of 100 or 1,000, you have one chance out of seven to be first, two chances out of seven to be second, and four chances out seven to be last in the market sequence;
- if you are proposed to buy at a price of 10,000, you have one chance out of four to be first, one chance out of four to be second, and two chances out four to be last in the market sequence;
- if you are proposed to buy at a price of 100,000, you have one chance out of two to be second, and one chance out of two to be third; in this case, you are sure not to be first in the market sequence;
- if you are proposed to buy at a price of 1,000,000, you are sure to be last in the market sequence.

Outcomes and final gain

You are going to participate in 10 independent periods of the game. The composition of each group is anonymous, randomly determined and different at each period of the game. After each period, once decisions are made, we will indicate (anonymously) if you were proposed to buy the asset and at what price. We will also indicate your gain for this period and your overall gain.

At the end of the ten periods of the game, we will compute your total gain as the sum of the gains in each period. Your total gain will be paid to you in euros. The rate of exchange is: 1 euro for 1 ECU.

Do you have any question?

QUESTIONNAIRE

- Is it possible to be first and be proposed to buy at a price of 100.000?
  - Yes
  - No
- If you are proposed to buy at a price of 1, are you sure to be first?
  - Yes
  - No
- If you are proposed to buy at a price of 100.000, are you sure to be second?
  - Yes
  - No
- If you are proposed to buy at a price of 100, are you sure to be third?
  - Yes
  - No
- If you are proposed to buy at a price of 1.000.000, are you sure to be third?
  - Yes
  - No
- If you accept to buy at 100, you will propose to resell at
  - 1.000
  - 1.100
- If you are first, buy at 100, and resell at 1,000, your payoff in the game is
  - 10 euros
  - 900 euros
- If you are first, buy at 100, and find nobody to resell to at 1,000, your payoff in the game is
  - 0 euro
  - 10 euros
- If you refuse to buy, your payoff is
  - 1 euro
  - 0 euro

Appendix B. Proofs of the analytical results

Notations

In all the proofs below, 10^K, where $K \in \{0, 4\}$, is the cap on the initial price (i.e., the highest potential price is $P = 10^{K+2}$), $a_i(t)$ is the action chosen by agent i at period t, the set of potential actions j is $\{B, \varnothing\}$ where B stands for a decision to buy, $\varnothing$ for a decision not to buy, $A_i^j(t+1|P)$ is the attraction of action j at period $t+1$ for agent i updated as shown in Eq. (2), and $Pr_i^j(t+1|P)$ is the probability for agent i to choose action j at period $t+1$ defined in Eq. (3). Denote $i-$ and $i+$ as the previous and next trader for trader i.

In all the proofs below, we exclude the case in which $\lambda = 0$. In this case indeed, the expected probability to buy is constant and equal to $\frac{1}{2}$ regardless of the price, the period and the past experience.

Proposition 1

Proof. To prove that the trading game converges to the no bubble equilibrium when $\lambda > 0$, we first show that no bubble arises conditional on the highest price. Then we show the same is true for the second highest price. Lastly, we apply the same logic to other lower prices and argue that no bubble arises conditional on those prices too.

We want to show that for any trader i, his probability to buy given any price P will converge to 0 in the long run. □

From Eq. (3), the probability to buy of trader i given price P at period $t+1$ is determined by the difference between his attraction for the action not to buy and that of his action to buy conditional on price P at period t, i.e., $A_i^{\varnothing}(t|P) - A_i^{B}(t|P)$.

We have:

$$A_i^{\varnothing}(t|P) - A_i^B(t|P) = A_i^{\varnothing}(t-1|P) - A_i^B(t-1|P) + \Delta\pi_i(t|P), \tag{4}$$

where we define $\Delta\pi_i(t|P)$ as the incremental payoff between no buy and buy. Computation yields:

$$\Delta\pi_i(t|P) = \begin{cases} -\left[\delta + (1-\delta)\mathbb{1}_{B=a_i(t)}\right]\pi(B, a_{-i}(t)) & \text{if condition } \mathbb{A} \text{ satisfies} \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

Whether the probability to buy of trader i would increase or decrease between t and $t+1$ thus depends on $\Delta\pi_i(t|P)$.

- First, we show that no bubble arises on the highest price, i.e., $P = 10^{K+2}$, in the long run. Given that the trader is proposed the highest price, he will never be able to resell the asset, thus $\pi(B, a_{-i}(t)) = -1$. It follows that $\Delta\pi_i(t|10^{K+2})$ can only take three values, 0, δ or 1:

$$\Delta\pi_i(t|10^{K+2}) = \begin{cases} 1 & \text{i is proposed price } 10^{K+2} \text{ and chooses to buy} \\ \delta & \text{i is proposed price } 10^{K+2} \text{ and chooses not to buy} \\ 0 & \text{i is not proposed price } 10^{K+2}. \end{cases} \tag{6}$$

  Iterating Eq. (4) from period 1 to t, and using our assumption that $A_i^{\varnothing}(0|P) = A_i^B(0|P) = 0$ to initialize the series, the probability to buy at period $t+1$ conditional on the highest price for trader i can be written as:

$$Pr_i^B(t+1|10^{K+2}) = \frac{1}{1 + e^{\lambda \sum_{\tau=1}^{t} \Delta\pi_i(\tau|10^{K+2})}}. \tag{7}$$

  From the discussion above, we have $\Delta\pi_i(t|10^{K+2}) \geq 0$.
  - Case I: the probability for trader i observing price 10^{K+2} does not converge to 0 as $t \to +\infty$, i.e., the probability to buy with second and third largest price does not converge to 0. In this case, we suppose that the probability to buy conditional on the highest price does not converge to 0 either. It implies that $\Delta\pi_i(t|10^{K+2}) > 0$ always holds at each period with a non-negligible probability. Thus, $plim_{t\to\infty} \sum_{\tau=1}^{t} \Delta\pi_i(\tau|10^{K+2}) = +\infty$ and $plim_{t\to\infty} Pr_i^B(t+1|10^{K+2}) = 0$. It contradicts the initial assumption. Thus, the probability to buy given the highest price will converge to 0 as $t \to +\infty$.
  - Case II: the probability for trader i observing price 10^{K+2} converges to 0 when $t \to +\infty$, i.e., the probability to buy with second or third largest price converges to 0. In this case, trading already stops at the second or third largest price. Thus, no bubble arises with the highest price either.
- Second, we show that no bubble arises on the second highest price, i.e., $P = 10^{K+1}$, in the long run.
  - First, we consider the case where the probability to buy given the highest prices converge to 0 in the long run. That is, $\forall\epsilon > 0$, $\zeta > 0$, there exists a T, such that $t > T$, $Pr(\left|Pr_i^B(t|10^{K+2}) - 0\right| > \epsilon) < \zeta$. We consider the case where $t \geq T$, the probability to buy conditional on the second highest price for any trader j is

$$Pr_j^B(t+1|10^{K+1}) = \frac{1}{1 + e^{-\lambda A_j^B(T-1|10^{K+1})} e^{\lambda \sum_{\tau=T}^{t} \Delta\pi_i(\tau|10^{K+1})}}, \tag{8}$$

    where $e^{-\lambda A_j^B(T-1|10^{K+1})}$ is bounded and $\Delta\pi_i(t|10^{K+1})$ are as follows:

$$\Delta\pi_i(t|10^{K+1}) = \begin{cases} 1 & \text{i is proposed price } 10^{K+1} \text{ and buys but cannot sell} \\ -9 & \text{i is proposed price } 10^{K+1} \text{ and buys and can sell} \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

    - Case I where the probability for trader j observing price 10^{K+1} does not converge to 0 as $t \to +\infty$, i.e., the probability to buy with third or fourth largest price does not converge to 0. In this case, we suppose that the probability to buy conditional on the second highest price does not converge to 0 either. It implies that $\Delta\pi_i(t|10^{K+1}) > 0$ happens at each period with a non-negligible probability.
      Since $\forall\epsilon > 0$, $\zeta > 0$, there exists a T, such that $t > T$, $Pr(\left|Pr_i^B(t|10^{K+2}) - 0\right| > \epsilon) < \zeta$. Thus, if trader j is proposed price 10^{K+1}, his probability to sell, i.e., the probability that $\Delta\pi_i(t|10^{K+1}) = -9$, converges to 0 as t is sufficiently large. In summary, the probability that $\Delta\pi_i(t|10^{K+1}) \geq 0$ approaches to 1 and $\Delta\pi_i(t|10^{K+1}) > 0$ happens with a non-negligible probability at each period when t is sufficiently large. Thus, $plim_{t\to\infty} \sum_{\tau=T}^{t} \Delta\pi_i(\tau|10^{K+1}) = +\infty$. Since $e^{-\lambda A_j^B(T-1|10^{K+1})}$ is bounded, we obtain that $plim_{t\to\infty} \mathrm{Pr}_i^B(t+1|10^{K+1}) = 0$. It contradicts the initial assumption. Thus, the probability to buy given the second highest price converges to 0 in the long run.
    - Case II where the probability for trader j observing price 10^{K+1} converges to 0 when $t \to +\infty$, i.e., the probability to buy with third or fourth largest price converges to 0. In this case, trading already stops at the third or second largest price. Thus, no bubble arises with the second highest price either.
  - Second, we consider the case where no bubble arises for the highest price because that trading already stops at the second or third largest price. It is obvious that, in this case, no bubble arises with the second highest price either.
- Last, according to the same reasoning, the same analysis can be applied to the traders with other lower prices and we can show that no bubble arises conditional on those prices. Therefore, in the long run, the trading should converge to no bubble equilibrium.

Proposition 2

Proof. In order to prove Proposition 2, we calculate the expected probabilities to buy at periods 1 and 2 conditional on each possible price. Then, we compare the expected probabilities to buy at these two different periods for different prices, to show how probability to buy would evolve from the first to the second period of the trading game.

At period 1, the probability to buy for any trader at any price is $\frac{1}{2}$ since the initial attractions are assumed to be 0. Let us analyze whether the probability to buy of a trader i at period 2 for any price $P \in \{1, \ldots, 10^k, \ldots, 10^{K+2}\}$ is higher or smaller than $\frac{1}{2}$. □

Let us denote by q_r^P the probability for a trader who is in position $r \in \{1, 2, 3\}$ in the market sequence to observe price P. More precisely, $q_1^P = Pr(\text{First observes } P)$, $q_2^P = Pr(\text{Second observes } P)$, and $q_3^P = Pr(\text{Third observes } P)$. We have:

$$q_1^{10^k} = q_2^{10^{k+1}} = q_3^{10^{k+2}} = \begin{cases} \frac{1}{2}^{k+1} & \text{if } k < K \\ \frac{1}{2}^{K} & \text{if } k = K \\ 0 & \text{if } k > K. \end{cases} \tag{10}$$

At period 1, there are four different potential cases. Trader i will observe price P with probability $\frac{1}{3}(q_1^P + q_2^P + q_3^P)$. His incremental payoff between not buy and buy thus writes as follows:

- With probability $Pr(\text{First observes } P \bigcap i \text{ is First}) = \frac{1}{3}q_1^P$, trader i observes price P and is in first position. In this case,

$$\Delta\pi_i(1|P) = \begin{cases} -9 & \text{if } a_i(1) = B \text{ and } a_{i+}(1) = B \quad \left(\frac{1}{4}\right) \\ 1 & \text{if } a_i(1) = B \text{ and } a_{i+}(1) = \varnothing \quad \left(\frac{1}{4}\right) \\ 0 & \text{if } a_i(1) = \varnothing \quad \left(\frac{1}{2}\right), \end{cases} \tag{11}$$

  where the number in the bracket represents the probability of each occasion. If trader i buys the asset and the next trader also buys the asset, which happens with probability $\frac{1}{4}$, the incremental payoff the trader i can obtain between not buy and buy is -9. If trader i buys the asset but the next trader refuses to buy, which happens with probability $\frac{1}{4}$, the incremental payoff is 1. Finally, if trader i refuses to buy, which happens with probability $\frac{1}{2}$, the incremental payoff is 0. In the latter case, imagination for what would happen if he had bought would not kick in since trader i is not provided the information on the next trader.
- With probability $Pr(\text{Second observes } P \bigcap i \text{ is Second}) = \frac{1}{3}q_2^P$, trader i is proposed price P and is in second position. This yields:

$$\Delta\pi_i(1|P) = \begin{cases} 0 & \text{if } a_{i-}(1) = \varnothing \quad \left(\frac{1}{2}\right) \\ -9 & \text{if } a_{i-}(1) = B \text{ and } a_i(1) = B \text{ and } a_{i+}(1) = B \quad \left(\frac{1}{8}\right) \\ 1 & \text{if } a_{i-}(1) = B \text{ and } a_i(1) = B \text{ and } a_{i+}(1) = \varnothing \quad \left(\frac{1}{8}\right) \\ 0 & \text{if } a_{i-}(1) = B \text{ and } a_i(1) = \varnothing \quad \left(\frac{1}{4}\right). \end{cases} \tag{12}$$

  In this case, the incremental payoff not only depends on the action of the next trader, but also on that of the previous one. If the previous trader chooses not to buy, which happens with probability $\frac{1}{2}$, the incremental payoff is 0; otherwise, the three cases are similar to those described in Eq. (11).
- With probability $Pr(\text{Third observes } P \bigcap i \text{ is Third}) = \frac{1}{3}q_3^P$, trader i is proposed price P and is in third position. Following the same reasoning as above, if $P < 10^{K+2}$, we have:

$$\Delta\pi_i(1|P) = \begin{cases} 1 & \text{if } a_i(1) = B \text{ and } a_{i-}(1) = B \text{ and } a_{i--}(1) = B \quad \left(\frac{1}{8}\right) \\ 0 & \text{otherwise} \quad \left(\frac{7}{8}\right), \end{cases} \tag{13}$$

  where $a_{i--}(1)$ is the action of the trader with two positions before trader i. In this case, the incremental payoff depends on not only the action of trader i but also the actions of the previous two traders. If the previous two traders and trader i choose to buy, which happens with probability $\frac{1}{8}$, the incremental payoff is 1; otherwise, the incremental payoff is 0.

The case in which trader i is proposed the highest possible price, i.e., $P = 10^{K+2}$ is slightly different: in that case indeed, a trader who would not buy could still imagine the payoff he would have received if he had bought given that he infers from P his position in the market sequence. This yields:

$$\Delta\pi_i(1|10^{K+2}) = \begin{cases} 1 & \text{if } a_i(1) = B \text{ and } a_{i_-}(1) = B \text{ and } a_{i_{--}}(1) = B \quad \left(\frac{1}{8}\right) \\ \delta & \text{if} a_i(1) = \varnothing \text{ and } a_{i_-}(1) = B \text{ and } a_{i_{--}}(1) = B \quad \left(\frac{1}{8}\right) \\ 0 & \text{otherwise} \quad \left(\frac{3}{4}\right). \end{cases} \tag{14}$$

In this case, the incremental payoff is 1 if all the three traders choose to buy, and δ if the first two traders choose to buy and trader i chooses not to buy since he can imagine that he would have received -1 if he had bought. For other cases, the incremental payoff is 0.

- With probability $1 - \frac{1}{3}\left(q_1^P + q_2^P + q_3^P\right)$, trader i is not proposed price P and $\Delta\pi_i(1|P) = 0$.

To sum up, the incremental payoff trader i can obtain between not buy and buy is as follows: for price $P < 10^{K+2}$,

$$\Delta\pi_i(1|P) = \begin{cases} 0 & \text{with prob. } 1 - \frac{1}{3}(q_1^P + q_2^P + q_3^P) + \frac{1}{6}\left(q_1^P + \frac{3}{2}q_2^P + \frac{7}{4}q_3^P\right) \\ 1 & \text{with prob. } \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P + \frac{1}{2}q_3^P\right) \\ -9 & \text{with prob. } \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P\right), \end{cases} \tag{15}$$

and for the highest possible price $P = 10^{K+2}$,

$$\Delta\pi_i(1|10^{K+2}) = \begin{cases} 0 & \text{with prob. } 1 - \frac{1}{3}(q_1^P + q_2^P + q_3^P) + \frac{1}{6}\left(q_1^P + \frac{3}{2}q_2^P + \frac{3}{2}q_3^P\right) \\ 1 & \text{with prob. } \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P + \frac{1}{2}q_3^P\right) \\ -9 & \text{with prob. } \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P\right) \\ \delta & \text{with prob. } \frac{1}{24}q_3^P. \end{cases} \tag{16}$$

We now compare the probability to buy at period 2 with the probability to buy at period 1 which is $\frac{1}{2}$. From Eq. (3), the probability to buy for trader i conditional on price P at period 2 can be written as:

$$Pr_i^B(2|P) = \frac{1}{1 + e^{\lambda\Delta\pi_i(1|P)}}, \tag{17}$$

where $\Delta\pi_i(1|P)$ is distributed according to Eq. (15) if $P < 10^{K+2}$ and Eq. (16) if $P = 10^{K+2}$.

Denote $E_1(Pr_i^B(2|P))$ as the expected probability to buy at period 2 for trader i conditional on price P. We can obtain that, if $P < 10^{K+2}$,

$$\begin{aligned} E_1(Pr_i^B(2|P)) = &\left[1 - \frac{1}{3}(q_1^P + q_2^P + q_3^P) + \frac{1}{6}\left(q_1^P + \frac{3}{2}q_2^P + \frac{7}{4}q_3^P\right)\right] \times \frac{1}{2} + \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P + \frac{1}{2}q_3^P\right) \times \frac{1}{1+e^{\lambda}} \\ &+ \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P\right) \times \frac{1}{1+e^{-9\lambda}}, \end{aligned} \tag{18}$$

and if $P = 10^{K+2}$,

$$\begin{aligned} E_1(Pr_i^B(2|10^{K+2})) = &\left[1 - \frac{1}{3}(q_1^P + q_2^P + q_3^P) + \frac{1}{6}\left(q_1^P + \frac{3}{2}q_2^P + \frac{3}{2}q_3^P\right)\right] \times \frac{1}{2} + \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P + \frac{1}{2}q_3^P\right) \times \frac{1}{1+e^{\lambda}} \\ &+ \frac{1}{12}\left(q_1^P + \frac{1}{2}q_2^P\right) \times \frac{1}{1+e^{-9\lambda}} + \frac{1}{24}q_3^P\frac{1}{1+e^{\lambda\delta}}. \end{aligned} \tag{19}$$

Finally, the expected probability at period 2 for trader i conditional on price P can be simplified as follows:

$$E_1(Pr_i^B(2|P)) = \begin{cases} \frac{1}{12}\left[6 - q_1^P - \frac{q_2^P}{2} - \frac{q_3^P}{4} + \frac{q_1^P + \frac{q_2^P}{2}}{1+e^{-9\lambda}} + \frac{2q_1^P + q_2^P + q_3^P}{2(1+e^{\lambda})}\right] & \text{if } P < 10^{K+2} \\ \frac{1}{24}\left[12 + (2q_1^P + q_2^P)\Psi_1 + q_3^P\Psi_2\right] & \text{if } P = 10^{K+2}, \end{cases} \tag{20}$$

Table A.1
Price Cap on the first price is 1.

Price P	q_1^P	q_2^P	q_3^P	$E_1(Pr_i^B(2\|P))$
1	1	0	0	$\frac{1}{12}\left(5+\frac{1}{1+e^{-9\lambda}}+\frac{1}{1+e^{\lambda}}\right)$
10	0	1	0	$\frac{1}{24}\left(11+\frac{1}{1+e^{-9\lambda}}+\frac{1}{1+e^{\lambda}}\right)$
100	0	0	1	$\frac{1}{24}\left(11+\frac{1}{1+e^{\lambda}}+\frac{1}{1+e^{\delta\lambda}}\right)$

Table A.2
Price Cap on the first price is 10^4.

Price P	q_1^P	q_2^P	q_3^P	$E_1(Pr_i^B(2\|P))$
1	$\frac{1}{2}$	0	0	$\frac{1}{24}\left(11+\frac{1}{1+e^{-9\lambda}}+\frac{1}{1+e^{\lambda}}\right)$
10	$\frac{1}{4}$	$\frac{1}{2}$	0	$\frac{1}{24}\left(11+\frac{1}{1+e^{-9\lambda}}+\frac{1}{1+e^{\lambda}}\right)$
10^2	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{24}\left(\frac{45}{4}+\frac{1}{2}\frac{1}{1+e^{-9\lambda}}+\frac{1}{1+e^{\lambda}}\right)$
10^3	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{192}\left(93+\frac{2}{1+e^{-9\lambda}}+\frac{4}{1+e^{\lambda}}\right)$
10^4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{384}\left(188+\frac{3}{1+e^{-9\lambda}}+\frac{5}{1+e^{\lambda}}\right)$
10^5	0	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{768}\left(381+\frac{2}{1+e^{-9\lambda}}+\frac{4}{1+e^{\lambda}}\right)$
10^6	0	0	$\frac{1}{16}$	$\frac{1}{384}\left(191+\frac{1}{1+e^{\lambda}}+\frac{1}{1+e^{\lambda\delta}}\right)$

where $\Psi_1 = \frac{1}{1+e^{\lambda}} + \frac{1}{1+e^{-9\lambda}} - 1$ and $\Psi_2 = \frac{1}{1+e^{\lambda}} + \frac{1}{1+e^{\delta\lambda}} - 1$.

It is useful to compute the probabilities q_r^P for $r \in \{1, 2, 3\}$ in Eq. (10) explicitly. Tables A.1 and A.2 show the values of these probabilities for any price P when the cap on the first price are 1 and 10^4 respectively. Notice that $0 \le q_1^P + q_2^P + q_3^P \le 1$.

- First, we analyze the case where trader i is proposed the highest price. Notice that $q_1^{10^{K+2}} = q_2^{10^{K+2}} = 0$ and $q_3^{10^{K+2}} > 0$. If trader i is proposed the highest price, we obtain from Eq. (20) that

$$E_1(Pr_i^B(2|10^{K+2})) = \frac{1}{24}\left[12 + q_3^{10^{K+2}}\left(\frac{1}{1+e^{\lambda}} + \frac{1}{1+e^{\delta\lambda}} - 1\right)\right],$$

  which is strictly lower than $\frac{1}{24}[12 + q_3^{10^{K+2}}(\frac{1}{2} + \frac{1}{2} - 1)] = \frac{1}{2}$ for $\lambda > 0$ and $\delta \ge 0$. The probability to buy of a trader who is proposed the highest price is expected to decrease immediately between period 1 and period 2.
- Second, let us analyze the cases where trader i is proposed the two lowest prices, $P = 1$ or $P = 10$. Notice that trader i can never be last, i.e., $q_3^1 = q_3^{10} = 0$. It follows from Eq. (20) that:

$$E_1(Pr_i^B(2|P)) = \frac{1}{12}\left[6 - q_1^P - \frac{1}{2}q_2^P + \left(q_1^P + \frac{q_2^P}{2}\right)\left(\frac{1}{1+e^{-9\lambda}} + \frac{1}{1+e^{\lambda}}\right)\right].$$

  Now, we have if $\lambda > 0$:

$$\frac{1}{1+e^{-9\lambda}} + \frac{1}{1+e^{\lambda}} = 1 - \frac{1}{1+e^{9\lambda}} + \frac{1}{1+e^{\lambda}} > 1 - \frac{1}{1+e^{\lambda}} + \frac{1}{1+e^{\lambda}} = 1.$$

  Based on the above two results, we can see that the expected probability to buy at period 2 when $P \in \{1, 10\}$ satisfies

$$E_1(Pr_i^B(2|P)) > \frac{1}{12}\left[6 - q_1^P - \frac{1}{2}q_2^P + \left(q_1^P + \frac{q_2^P}{2}\right)\right] = \frac{1}{2}.$$

  Therefore, the expected probability to buy increases for prices 1 and 10 at period 2 whatever the cap on the initial price.
- Third, when the cap is $K = 4$, let us analyze the cases where trader i is proposed the price is 10^k, where $k = 2, 3, 4, 5$. To this end, we differentiate the expected probability to buy defined in Eq. (20) with respect to λ, using the exact values of q_r^P defined in Table A.2. Computation yields:

$$\frac{\partial E_1(Pr_i^B(2|10^2))}{\partial\lambda} = \frac{1}{48}e^{\lambda}\left(\frac{9e^{8\lambda}}{(1+e^{9\lambda})^2} - \frac{2}{(1+e^{\lambda})^2}\right),$$
$$\frac{\partial E_1(Pr_i^B(2|10^3))}{\partial\lambda} = \frac{1}{96}e^{\lambda}\left(\frac{9e^{8\lambda}}{(1+e^{9\lambda})^2} - \frac{2}{(1+e^{\lambda})^2}\right),$$
$$\frac{\partial E_1(Pr_i^B(2|10^4))}{\partial\lambda} = \frac{1}{384}e^{\lambda}\left(\frac{27e^{8\lambda}}{(1+e^{9\lambda})^2} - \frac{5}{(1+e^{\lambda})^2}\right),$$
$$\frac{\partial E_1(Pr_i^B(2|10^5))}{\partial\lambda} = \frac{2}{768}e^{\lambda}\left(\frac{9e^{8\lambda}}{(1+e^{9\lambda})^2} - \frac{2}{(1+e^{\lambda})^2}\right).$$

It is easy to show that the first order differentiation is strictly positive if and only if $0 \leq \lambda < \lambda^*(P)$, where $\lambda^*(P) \in (0, +\infty)$. That is, for all $P \in \{10^2, 10^3, 10^4, 10^5\}$, there exists a $\lambda^*(P) \in (0, +\infty)$ such that the expected probability to buy at price P increases with λ when $\lambda \leq \lambda^*(P)$ and decreases with λ when $\lambda > \lambda^*(P)$.

When $\lambda = 0$, the expected probability to buy at period 2 is constant and equal to $\frac{1}{2}$ regardless of the price. When $\lambda \to +\infty$, the expected probability to buy at period 2 are $\frac{47}{96}$, $\frac{95}{192}$, $\frac{191}{384}$ and $\frac{383}{768}$ at prices 10^2, 10^3, 10^4 and 10^5 respectively. These probabilities are all lower than $\frac{1}{2}$. Therefore, there exists a $\lambda'(P) \in (0, +\infty)$ such that the expected probability to buy at period 2 is greater than $\frac{1}{2}$ if $\lambda \in (0, \lambda'(P))$ and lower than $\frac{1}{2}$ if $\lambda \in (\lambda'(P), +\infty)$.

Appendix C. Simulations when a trader observes subsequent trader's behavior

This appendix displays the results of simulations for the same game as in the experiment except that it is assumed that traders can observe the decision of the next trader in the market sequence. Fig. A.1 refers to the case in which the price cap is 1. Fig. A.2 refers to the case in which the price cap is 10,000.

When traders observe the actions of the subsequent trader, the propensity to speculate is initially stronger than with no imagination because imaginative traders who are proposed low prices realize that they could have obtained a higher payoff by buying the asset if subsequent traders decided to buy. The speed at which traders learn is then faster.

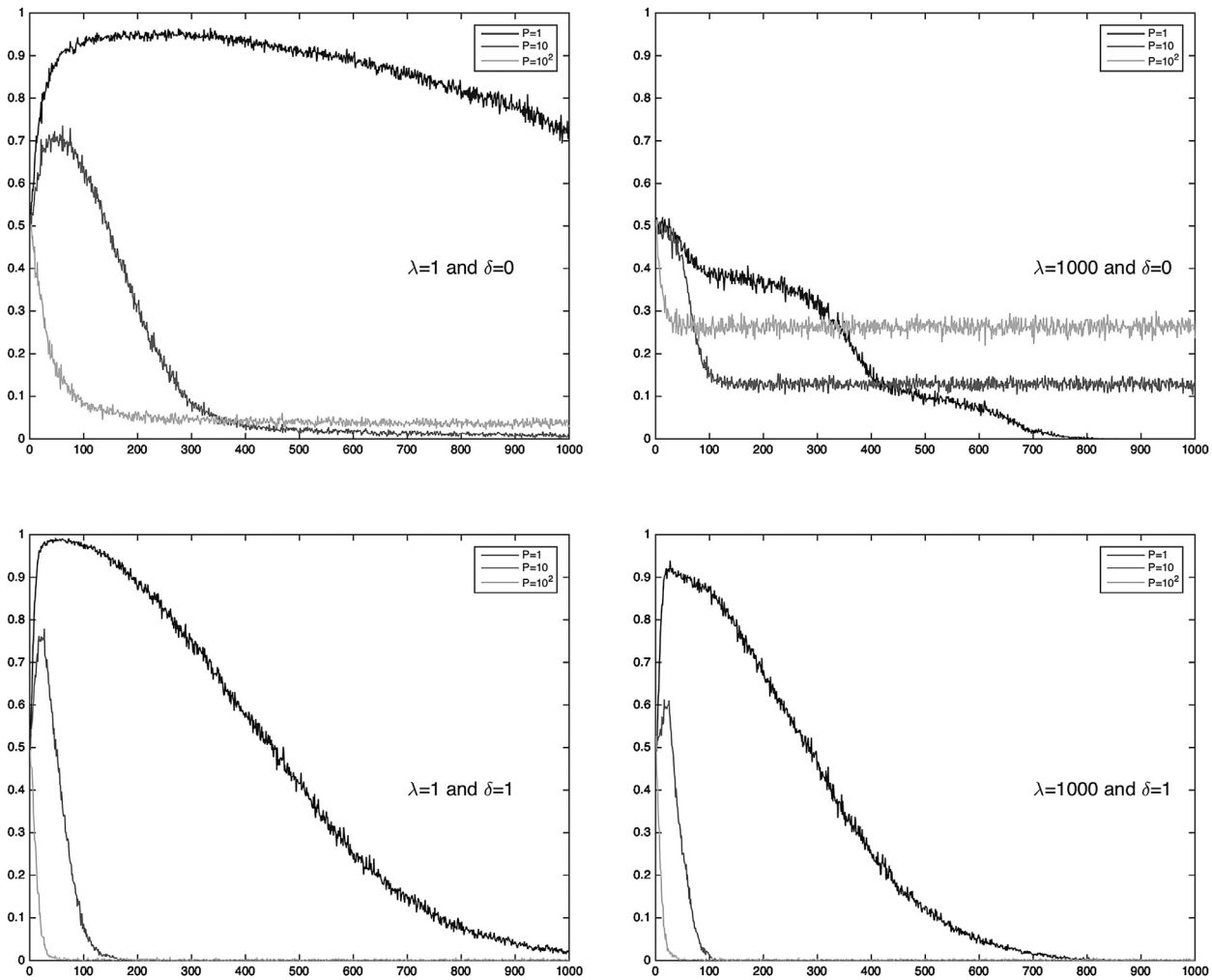


Fig. A.1. Probability to speculate per price in the EWA learning model when the cap on the first price is 1. Setting in which a trader observes subsequent trader's behavior. This figure displays the average probability to buy in the Bubble game assuming that traders can observe the choice of the next trader in the market sequence. Behavior is modeled as in the EWA learning model of Camerer and Ho (1999). We set $A_i^j(0|P) = 0$.

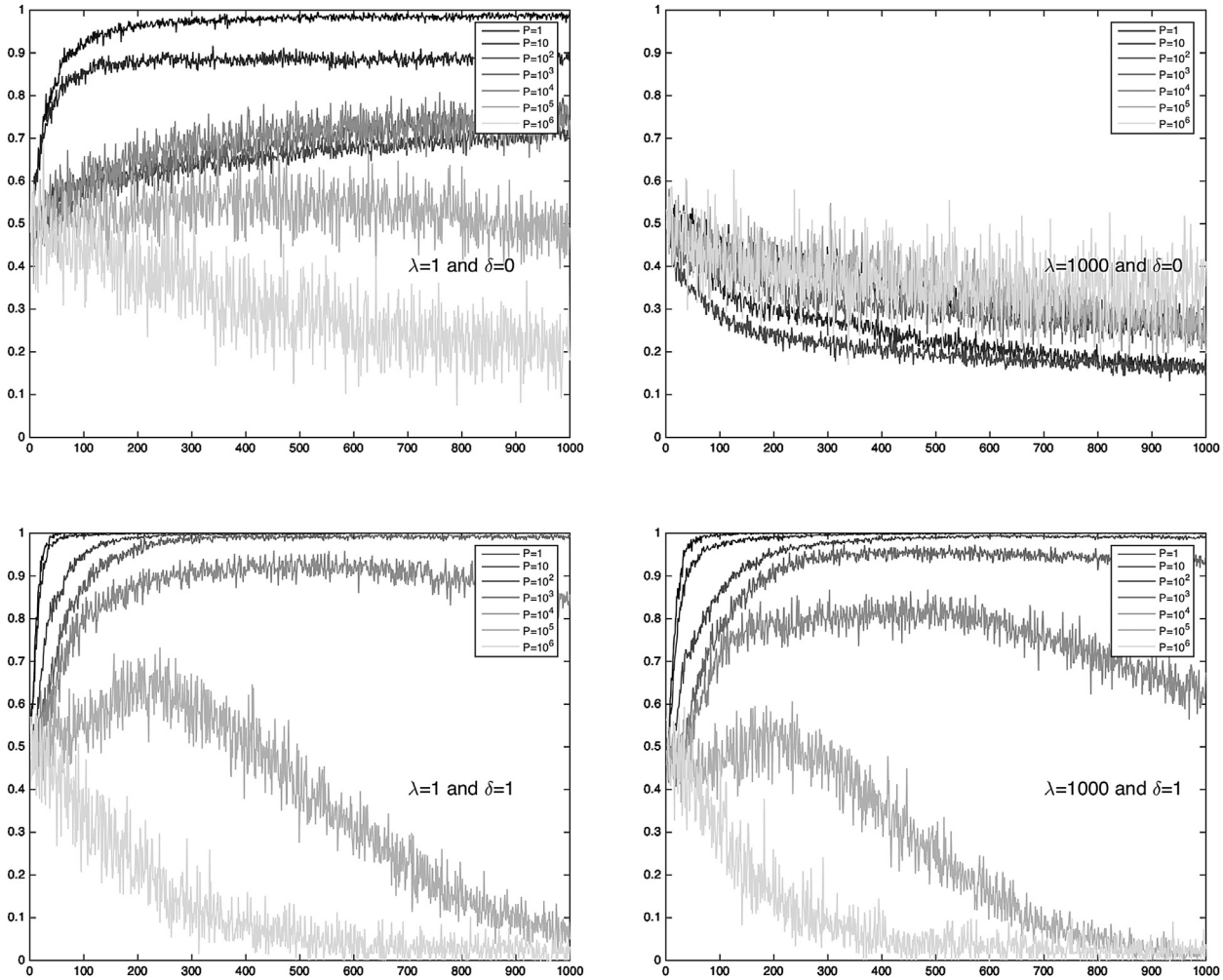


Fig. A.2. Probability to speculate per price in the EWA learning model when the cap on the first price is 10,000. Setting in which a trader observes subsequent trader's behavior. This figure displays the average probability to buy in the Bubble game assuming that traders can observe the choice of the next trader in the market sequence. Behavior is modeled as in the EWA learning model of Camerer and Ho (1999). We set $A_i^j(0|P) = 0$.

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